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Base Metals Supply & Demand Tracker

While market awaits tariff update, China keeps buying copper

Global Commodities Research

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Copper:

- **Chinese copper demand indicators slump vs a very high base effect last year…** Our consumption-weighted end-use indicator for Chinese copper demand declined sharply in May to now register at -5% yoy YTD through 5M26 (Figure 23). Significant weakness was maintained across construction while renewable installations faced a final month in May of an extremely elevated base level last year given the rush to install capacity before June 2025 when the tariff rate for wind and solar installations moved off the more advantageous fixed-price tariff system. Adding to the outsized yoy weakness in May, grid investment also came in well below a uniquely strong May 2025 base level, likely tied in with the massive renewable rush last year. Nonetheless, YTD grid investment is still around +13% yoy through May and the particularly strong base level effects in renewables and grid will largely fade from June onwards.
- **While apparent consumption continues to diverge higher amid resilient refined production and continued inventory draws.** Even when facing a similarly strong 2025 base, Chinese apparent consumption continued to diverge from underlying demand indicators registering 8% yoy growth in May (~+3% yoy YTD through 5M26) (Figure 24). Chinese refined copper production is still hovering at almost 14 mmt (Figure 43 & Figure 44), helped in part by solid scrap imports (Figure 41), despite a 1% yoy drop in copper concentrate imports YTD through May (Figure 38) . Amid this resilient production, refined copper imports came in at 316 kmt in May (close to 2025 average - Figure 39) while visible copper inventory continued to draw and has now fallen to the lows of the five year range (Figure 31).
- **Ex-US inventory levels come back into focus again amid Chinese stock draws and LME cancellations.** With Chinese SHFE + bonded stocks now drawing to sub 140 kmt, below the five year range, Chinese premiums have pushed up towards $95/mt, the highest since May 2025 and China's import arb has largely been open since the start of July (Figure 36). Amid this backdrop, LME cancellations for copper deliveries out of Asian warehouses have risen by more than 65 kmt so far this week, dropping overall on-warrant LME inventory globally back to only around 130 kmt, approaching critically low levels which can drive periods of sharp backwardation.
- **Market still awaiting an update following the review of Section 232.** In our view, the key 2H26 catalyst for copper is policy, not fundamentals: the structure/communication following the US review of Section 232 copper tariffs (*Copper: Where we're going, we don't need balances*, 21 June 2026). For LME copper prices, escalation is everything as the most important flip

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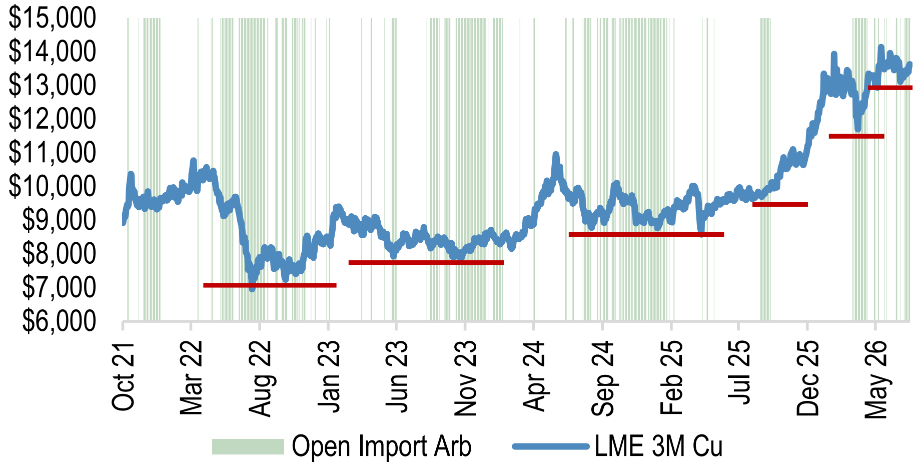
point between a bullish and bearish outcome is how any communication shapes expectations around the future rate of US import tariffs on refined copper cathode. While the US import arb remains open, monthly US imports of refined copper have faded off the highs of earlier this year (Figure 33& Figure 34). A solidification of an escalatory regime (our base case) equals an incentive to still import copper into the US before tariffs come into force or increase, which is bullish as this US pull on metal continues, China is forced to react (buy at increasingly higher levels to fill their import demand - Figure 1) and risks of reaching a very bullish pinch point driven by rapidly dwindling ex-US inventories remain very elevated. On the other hand, no expectations of escalation, even if an outright tariff is immediately announced, is bearish for LME prices as it will result in a closed US import arb window and a long de-stocking period in the US, giving China its pricing power back.

Figure 1: China's buying floor has moved sharply higher in the last year

LME 3M Copper prices (US$/mt) vs an open Chinese spot copper import arb window



Source: LME, Bloomberg Finance L.P., J.P. Morgan Commodities Research. *Arb as of 8:00 AM London.

Aluminium:

- **Middle East production starts its long path towards recovery.** EGA announced the restart of its alumina refinery last week, which supplies around 50% of the company's alumina needs, while its Al Taweelah smelter has begun its restart too. As of early July, frozen metal was removed from over 20% of reduction cells at the smelter and about 7% of the overall reduction cells had been restarted. This initial restart progress comes quicker than our initial expectations, though the company notes that hot metal production will ramp up gradually and could still take up to a year to reach pre-incident levels.
- **Chinese inventory draws finally come through on boosted exports but China's semis export arb has now closed again.** Since the end of May, Chinese visible inventories have drawn by around 320 kmt to around 1 mmt (Figure 56). This draw comes as boosted Chinese semis exports (incentivized by price differentials to backfill the supply gap left by the Middle East disruption) have eventually rolled up the supply chain to a result in a more material pull on Chinese aluminum ingot inventory (Figure 60). That all being said, as LME prices have sold off back to ~$3,150/mt, this export arb has closed again (Figure 63). Ultimately, even as Middle East supply recovers and Indonesian capacity expansions progress, we still see a sizeable 3Q26 ex-China deficit that will likely need to continue to pull on China's inventory, skewing risk towards another push higher in

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LME prices to once again re-open the export arb and bridge the supply gap until we reach a more well-supplied global market once again next year.

Zinc:

- **Waiting on China's export arb to open.** China continues to absorb displaced concentrate in the global market (Figure 83). Zinc concentrate imports have grown by 9% yoy over 5M26 and Chinese zinc smelter production remains broadly resilient, albeit weakening slightly in recent months (Figure 84), which illustrates the continued appetite of Chinese smelters even amid persistently weak demand onshore in China and more stressing margin pressure amid negative spot treatment charges (Figure 81). With galvaniser operating rates still below the 5-yr avg (Figure 87), onshore China zinc inventory continues to linger at recent historical highs (Figure 77). Ex-China, inventory coverage remains relatively low (Figure 76) and we still expect a more material opening in China's export arb (Figure 86) and subsequent outflow of metal from China in the coming months to alleviate this dislocation between ex-China and China zinc supply/ demand.

Nickel:

- **Focus on 2H26 RKAB revisions before the whole quota cycle starts again later this year.** After breaching above $19,000/mt LME nickel prices are now back to ~$17,000/mt following reports of a material boost to Indonesia's RKAB ore quota over the July revision window. Interestingly, realized ore and intermediary prices (NPI & nickel sulfate) have remained relatively steady during the same time, suggesting that the repricing in LME is more forward looking (Figure 91). Our Indonesia Metals and Mining team's initial assumption is for the government to increase RKAB permits by 30-40mn to 290-300mn wmt (vs 260-270mn earlier), largely for limonite as saprolite availability remains limited. Overall we still expect the Indonesian RKAB policy will remain active and flexible over the coming years to attempt to maintain balance and continue to defend and support a desired price outcome (not too high, not too low) and maximize tax revenue.

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Global demand indicators

Table 1: Global metals monthly price change heatmap

Percent change, MoM & YoY

	Global Metals Price Heatmap	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26	Jun-26	2023	2024	2025	2026 YTD
Metals Prices	LME Copper	3.9%	-2.6%	3.0%	3.7%	6.0%	2.8%	11.0%	5.9%	1.4%	-9.3%	5.3%	5.0%	5.0%	2.2%	2.4%	41.7%	10.2%
	LME Aluminium	6.3%	-1.3%	2.0%	2.5%	7.6%	-0.6%	4.4%	5.0%	-0.1%	3.8%	0.2%	5.5%	5.5%	0.3%	7.0%	17.4%	6.5%
	LME Zinc	5.0%	0.3%	2.1%	5.0%	3.2%	0.0%	2.0%	9.1%	-2.5%	-8.3%	4.2%	5.3%	5.3%	-10.6%	12.1%	4.7%	15.8%
	LME Nickel	-0.1%	-1.8%	3.2%	-1.2%	-0.1%	-2.6%	12.3%	7.9%	-0.6%	-5.0%	13.8%	-2.1%	-2.1%	-44.7%	-7.7%	8.6%	0.6%
	LME Lead	4.4%	-3.6%	1.0%	-0.1%	1.4%	-1.8%	1.5%	-0.1%	-2.3%	-3.5%	2.8%	3.1%	3.1%	-9.8%	-5.6%	3.0%	-7.7%
	LME Tin	10.9%	-3.0%	7.1%	1.1%	1.9%	8.5%	3.6%	28.1%	11.1%	-23.4%	5.3%	12.6%	12.6%	2.4%	14.4%	39.4%	32.8%
	LME Cobalt spot	-1.1%	0.0%	0.0%	5.1%	39.3%	0.0%	9.9%	5.5%	0.1%	0.0%	0.0%	0.0%	0.0%	-44.3%	-15.4%	117.8%	5.6%
	LME Lithium Hydroxide	-3.2%	-2.1%	5.4%	11.9%	1.0%	5.8%	10.1%	62.0%	-4.3%	12.5%	3.5%	7.8%	7.8%	-80.4%	-42.9%	18.5%	73.0%
	COMEX Copper	7.5%	-13.4%	3.8%	7.5%	4.8%	1.9%	9.6%	4.3%	1.4%	-9.7%	5.6%	7.8%	7.8%	2.1%	3.5%	41.1%	12.0%
	SHFE Copper	2.4%	-1.9%	1.0%	5.1%	4.8%	0.0%	13.1%	8.3%	-4.2%	-7.8%	5.8%	3.6%	3.6%	4.4%	6.8%	33.7%	5.8%
	SHFE Aluminium	2.8%	-0.5%	0.6%	0.0%	2.5%	1.1%	5.6%	10.6%	-6.0%	0.1%	-1.3%	-0.5%	-0.5%	4.5%	1.0%	14.7%	2.1%
	SHFE Zinc	-0.5%	-0.6%	-1.3%	-1.0%	2.0%	0.2%	4.1%	12.2%	-6.0%	-6.3%	0.4%	5.1%	5.1%	-9.5%	19.0%	-9.4%	6.1%
	Spot Gold	0.4%	-0.4%	4.8%	11.9%	3.7%	5.9%	1.9%	13.3%	7.9%	-15.2%	-1.1%	-1.7%	-1.7%	13.1%	27.2%	64.6%	-5.8%
	Spot Silver	9.5%	1.7%	8.2%	17.4%	4.4%	16.0%	26.8%	18.9%	10.1%	-24.1%	-1.9%	2.1%	2.1%	-0.7%	21.5%	148.0%	-17.9%
	Spot Platinum	28.5%	-5.0%	6.1%	14.9%	-0.1%	6.1%	23.3%	6.5%	7.9%	-18.5%	1.7%	-3.4%	-3.4%	-7.7%	-8.5%	127.0%	-20.4%
	Spot Palladium	13.6%	8.3%	-7.9%	14.3%	14.1%	1.1%	11.4%	5.7%	4.4%	-19.4%	3.6%	-11.2%	-11.2%	-38.6%	-17.1%	77.5%	-19.2%

*LME is 3-month rolling forward, except cobalt which is spot. Lithium hydroxide and SHFE prices are futures contracts. Prices as of 14th July 2026.
Source: Bloomberg Finance L.P.

Table 2: Global manufacturing PMI by country

		Jan-25	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26
Groups	**JP Morgan Global**	50.1	50.6	50.3	49.8	49.5	50.4	49.7	50.9	50.7	50.7	50.5	50.4	50.9	51.8	51.3	52.6	52.7
	Developed Markets	49.3	50.0	49.1	49.1	50.0	50.5	49.2	50.9	50.3	50.3	50.5	50.5	51.4	51.8	52.0	53.8	53.9
	Emerging Markets	50.8	51.1	51.3	50.5	49.2	50.4	50.1	50.9	51.2	51.2	50.5	50.4	50.6	52.0	50.7	51.6	51.6
	European Union	46.7	47.8	48.6	49.1	49.2	49.2	49.6	50.4	49.7	49.7	49.5	48.8	49.5	50.5	51.3	52.0	51.4
	Eurozone	46.6	47.6	48.6	49.0	49.4	49.5	49.8	50.7	49.8	49.8	49.6	48.8	49.5	50.8	51.6	52.2	51.6
Asia Pacific	China (Caixin/Markit)	50.1	50.8	51.2	50.4	48.3	50.4	49.5	50.5	51.2	51.2	49.9	50.1	50.3	52.1	50.8	52.2	51.8
	China (Official PMI)	49.1	50.2	50.5	49.0	49.5	49.7	49.3	49.4	49.8	49.8	49.2	50.1	49.3	49.0	50.4	50.3	50.0
	India	57.7	56.3	58.1	58.2	57.6	58.4	59.1	59.3	57.7	57.7	56.6	55.0	55.4	56.9	53.9	54.7	55.0
	Japan	48.7	49.0	48.4	48.7	49.4	50.1	48.9	49.7	48.5	48.5	48.7	50.0	51.5	53.0	51.6	55.1	54.5
Europe	France	45.0	45.8	48.5	48.7	49.8	48.1	48.2	50.4	48.2	48.2	47.8	50.7	51.2	50.1	50.0	52.8	49.7
	Germany	45.0	46.5	48.3	48.4	48.3	49.0	49.1	49.8	49.5	49.5	48.2	47.0	49.1	50.9	52.2	51.4	50.1
	Italy	46.3	47.4	46.6	49.3	49.2	48.4	49.8	50.4	49.0	49.0	50.6	47.9	48.1	50.6	51.3	52.1	52.9
	Spain	50.9	49.7	49.5	48.1	50.5	51.4	51.9	54.3	51.5	51.5	51.5	49.6	49.2	50.0	48.7	51.7	51.2
	United Kingdom	48.3	46.9	44.9	45.4	46.4	47.7	48.0	47.0	46.2	46.2	50.2	50.6	51.8	51.7	51.0	53.7	53.9
Americas	Canada	51.6	47.8	46.3	45.3	46.1	45.6	46.1	48.3	47.7	47.7	48.4	48.6	50.4	51.0	50.0	53.3	52.9
	Mexico	49.1	47.6	46.5	44.8	46.7	46.3	49.1	50.2	49.6	49.6	47.3	46.1	46.3	47.1	48.9	47.7	49.6
	United States	51.2	52.7	50.2	50.2	52.0	52.9	49.8	53.0	52.0	52.0	52.2	51.8	52.4	51.6	52.3	54.5	55.1
	US ISM	50.5	50.0	48.9	48.8	48.6	49.0	48.4	48.9	48.9	48.9	48.0	47.9	52.6	52.4	52.7	52.7	54.0

Source: Bloomberg Finance L.P.

Table 3: Global light vehicle production growth

Percent change

QoQ	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27	2024	2025	2026E	2027F
Europe	0.5%	2.4%	-13.4%	12.0%	1.5%	-1.7%	-12.3%	11.6%	0.2%	3.9%	-12.1%	10%				
China	-24.5%	8.0%	7.8%	17.1%	-32.2%	13.2%	8.9%	15.2%	-25.9%	17.1%	-8.2%	26%				
Japan/Korea	-5.4%	-0.4%	-3.3%	6.0%	0.0%	-3.3%	-4.9%	-0.9%	-0.4%	1.4%	-3.5%	6%				
North America	4.2%	5.0%	0.3%	-9.9%	4.0%	4.8%	-0.8%	-9.3%	3.9%	6.9%	-3.3%	-2%				
Other	4.4%	-4.9%	6.9%	1.1%	0.2%	-6.3%	2.0%	-3.0%	3.9%	4.5%	5.1%	-3%				
Global	**-8.9%**	**3.0%**	**0.8%**	**7.4%**	**-11.8%**	**2.9%**	**0.2%**	**5.3%**	**-9.1%**	**8.5%**	**-5.2%**	**10%**				
YoY																
Europe	-5.3%	-0.3%	3.8%	-0.2%	0.8%	-3.3%	-2.0%	-2.4%	-3.6%	1.9%	2.1%	0.3%	-4.4%	-0.7%	-1.7%	0.1%
China	14.5%	11.0%	14.6%	2.8%	-7.6%	-3.1%	-2.1%	-3.6%	5.2%	8.9%	-8.2%	0.4%	3.6%	10.0%	-4.0%	1.2%
Japan/Korea	7.0%	0.3%	0.8%	-3.5%	2.1%	-0.9%	-2.6%	-8.9%	-9.2%	-4.8%	-3.4%	3.0%	-6.4%	1.0%	-2.6%	-3.7%
North America	-5.0%	-3.5%	5.2%	-1.1%	-1.3%	-1.4%	-2.5%	-1.9%	-2.0%	-0.1%	-2.6%	5.8%	-1.5%	-1.2%	-1.8%	0.1%
Other	4.3%	5.6%	1.4%	7.3%	3.0%	1.5%	-3.1%	-7.0%	-3.6%	7.5%	10.7%	11.0%	-0.5%	4.6%	-1.5%	6.2%
Global	**3.9%**	**3.7%**	**6.9%**	**1.6%**	**-1.7%**	**-1.8%**	**-2.4%**	**-4.4%**	**-1.5%**	**3.9%**	**-1.7%**	**3.2%**	**-1.0%**	**3.9%**	**-2.6%**	**1.0%**

*3Q26-4Q27 is a forecast as of June 2026.
Source: S&P Global

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Figure 2: Vehicle sales and penetration rates

Vehicles in thousands

Global PV Sales ('000s)	Jun-26	Jun-25	y/y%	6M26	6M25	y/y%
Passenger Vehicles (PV)						
Europe (EU5)	973	868	12.1%	5,066	4,752	6.6%
Germany	296	256	15.7%	1,484	1,403	5.8%
France	189	170	11.4%	857	842	1.8%
Italy	146	132	10.8%	938	855	9.8%
Spain	128	119	7.8%	648	610	6.2%
United Kingdom	213	191	11.4%	1,138	1,042	9.2%
China	1,602	2,085	-23.2%	8,721	10,895	-20.0%
US	1,379	1,280	7.7%	7,954	8,181	-2.8%
Total	**3,954**	**4,233**	-6.6%	**21,740**	**23,828**	-8.8%
Battery Electric Vehicles (EV)						
Europe (EU5)	233	143	63.1%	1,037	713	45.4%
Germany	84	47	78.2%	368	249	47.9%
France	56	29	91.7%	242	149	62.5%
Italy	15	8	88.3%	80	45	77.8%
Spain	14	11	26.5%	63	46	36.6%
United Kingdom	64	47	35.0%	285	225	26.6%
China	685	661	3.6%	3,106	3,336	-6.9%
US	73	101	-27.6%	458	616	-25.7%
Total	**991**	**905**	9.6%	**4,601**	**4,665**	-1.4%
Plug-in Hybrid Vehicles (PHEV)						
Europe (EU5)	102	82	25.2%	516	394	31.1%
Germany	32	26	25.8%	164	139	17.8%
France	12	12	4.4%	42	49	-15.4%
Italy	16	10	64.6%	85	43	98.6%
Spain	16	14	15.0%	78	56	39.2%
United Kingdom	27	21	24.9%	148	107	38.4%
China	323	450	-28.2%	1,614	2,128	-24.2%
US	18	21	-15.9%	86	175	-51.1%
Total	**443**	**553**	-19.8%	**2,216**	**2,697**	-17.8%

Global EV/PHEV Penetration (%)	Jun-26	Jun-25	y/y%	6M26	6M25	y/y%
EV/PHEV Penetration (%)						
Europe (EU5)	34.5%	25.9%	8.6%p	30.7%	23.3%	7.4%p
Germany	39.2%	28.4%	10.8%p	35.8%	27.6%	8.2%p
France	36.1%	24.1%	12.0%p	33.0%	23.5%	9.5%p
Italy	20.9%	13.2%	7.7%p	17.5%	10.2%	7.3%p
Spain	23.2%	20.8%	2.4%p	21.8%	16.8%	5.0%p
United Kingdom	42.5%	35.9%	6.6%p	38.0%	31.8%	6.2%p
China	62.9%	53.3%	9.6%p	54.1%	50.1%	4.0%p
US	6.6%	9.5%	-2.9%p	6.8%	9.7%	-2.8%p
Total	**36.3%**	**34.4%**	1.8%p	**31.4%**	**30.9%**	0.5%p
EV Penetration (%)						
Europe (EU5)	23.9%	16.4%	7.5%p	20.5%	15.0%	5.5%p
Germany	28.4%	18.4%	10.0%p	24.8%	17.7%	7.1%p
France	29.6%	17.2%	12.4%p	28.2%	17.6%	10.5%p
Italy	10.2%	6.0%	4.2%p	8.5%	5.2%	3.2%p
Spain	11.1%	9.4%	1.6%p	9.8%	7.6%	2.2%p
United Kingdom	30.0%	24.8%	5.2%p	25.0%	21.6%	3.4%p
China	42.8%	31.7%	11.1%p	35.6%	30.6%	5.0%p
US	5.3%	7.9%	-2.6%p	5.8%	7.5%	-1.8%p
Total	**25.1%**	**21.4%**	3.7%p	**21.2%**	**19.6%**	1.6%p
PHEV Penetration (%)						
Europe (EU5)	10.5%	9.4%	1.1%p	10.2%	8.3%	1.9%p
Germany	10.9%	10.0%	0.9%p	11.0%	9.9%	1.1%p
France	6.5%	7.0%	-0.4%p	4.8%	5.8%	-1.0%p
Italy	10.7%	7.2%	3.5%p	9.1%	5.0%	4.0%p
Spain	12.1%	11.4%	0.8%p	12.0%	9.2%	2.9%p
United Kingdom	12.5%	11.2%	1.4%p	13.0%	10.3%	2.7%p
China	20.2%	21.6%	-1.4%p	18.5%	19.5%	-1.0%p
US	1.3%	1.6%	-0.4%p	1.1%	2.1%	-1.1%p
Total	**11.2%**	**13.1%**	-1.9%p	**10.2%**	**11.3%**	-1.1%p

*As of May 2026.
Source: CNEV Post, European Union, US Motor Intelligence, J.P. Morgan. Note: China data are based on preliminary CPCA retail sales data.

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China demand indicators

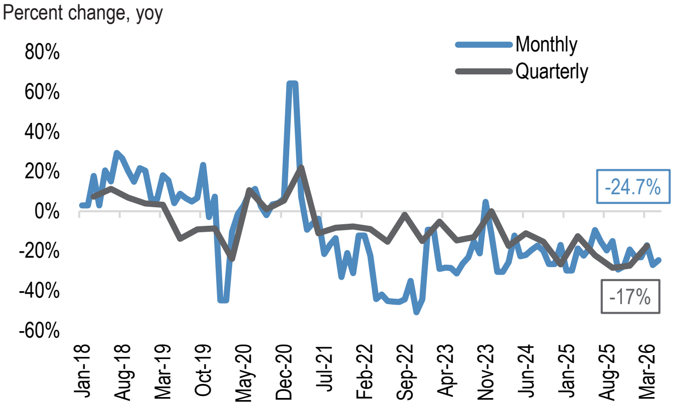
Table 4: China metals demand indicator heat map

	China End-Use Demand Indicators Heatmap	Units	Jan-23	Feb-23	Mar-23	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26
Property	70 cities newly built price index	YoY	-2.3%	-1.9%	-1.4%	-4.6%	-4.1%	-3.7%	-3.4%	-3.0%	-2.7%	-2.6%	-2.8%	-3.1%	-3.3%	-3.5%	-3.6%	-3.7%	-3.6%
	Residential building starts (million sqm)	YTD YoY	-8.7%	-8.7%	-17.8%	-22.3%	-21.4%	-19.6%	-18.3%	-18.3%	-18.3%	-19.3%	-19.9%	-19.8%	-23.3%	-23.3%	-22.0%	-23.6%	-23.4%
	Residential area under const. (million sqm)	YTD YoY	-4.7%	-4.7%	-5.4%	-10.1%	-9.6%	-9.4%	-9.4%	-9.6%	-9.7%	-9.7%	-10.0%	-10.3%	-11.9%	-11.9%	-12.1%	-12.5%	-12.6%
	Residential const. completed (m. sqm)	YTD YoY	9.7%	9.7%	16.8%	-16.8%	-17.6%	-15.5%	-17.3%	-18.5%	-17.1%	-18.9%	-20.1%	-20.2%	-26.9%	-26.9%	-26.5%	-25.8%	-25.0%
Infrastructure	FAI - total (RMB)	YTD YoY	5.5%	5.5%	5.1%	4.0%	3.7%	2.8%	1.6%	0.5%	-0.5%	-1.7%	-2.6%	-3.8%	1.8%	1.8%	1.7%	-1.6%	-4.1%
	FAI- manufacturing (RMB)	YTD YoY	8.1%	8.1%	7.0%	8.8%	8.5%	7.5%	6.2%	5.1%	4.0%	2.7%	1.9%	0.6%	3.1%	3.1%	4.1%	1.2%	-0.4%
	FAI- electric equipment and machinery (RMB)	YTD YoY	33.7%	33.7%	43.1%	-7.5%	-8.6%	-7.8%	-8.7%	-8.8%	-9.5%	-9.4%	-9.5%	-10.3%	5.8%	5.8%	0.8%	2.9%	1.5%
	FAI- railway transportation (RMB)	YTD YoY	17.8%	17.8%	17.6%	1.6%	2.3%	4.2%	5.9%	4.5%	4.2%	3.0%	2.7%	-1.2%	28.7%	28.7%	12.6%	8.6%	5.1%
	FAI- construction (RMB)	YTD YoY	71.9%	71.9%	32.1%	-0.3%	-19.8%	-21.2%	-24.7%	-24.7%	-17.2%	-13.7%	-14.7%	-22.2%	2.2%	2.2%	0.1%	-11.7%	-6.0%
	FAI- real estate (RMB)	YTD YoY	-5.1%	-5.1%	-5.1%	-10.6%	-11.1%	-11.5%	-12.4%	-13.2%	-14.0%	-14.9%	-16.0%	-17.5%	-10.7%	-10.7%	-10.6%	-13.4%	-15.9%
	FAI- investment in grid infrastructure (RMB)	YTD YoY	1.9%	1.9%	7.6%	14.6%	19.8%	14.6%	12.5%	14.0%	9.9%	7.2%	5.9%	5.1%	-4.0%	-4.0%	43.4%	32.4%	13.2%
	Cement production (million tonnes)	YTD YoY	-0.6%	-0.6%	10.4%	-5.3%	-8.1%	-5.3%	-5.6%	-6.2%	-8.6%	-15.8%	-8.2%	-6.6%	6.8%	6.8%	-21.0%	-10.8%	-8.1%
Manufacturing	China NBS Manufacturing PMI	Index	50.1	52.6	51.9	49.0	49.5	49.7	49.3	49.4	49.8	49.0	49.2	50.1	49.3	49.0	50.4	50.3	50.0
	Refrigerator Production (units)	YTD YoY	10.6%	10.6%	7.8%	-2.0%	-2.6%	0.3%	1.7%	3.5%	3.8%	3.1%	3.6%	5.1%	10.1%	10.1%	12.7%	15.9%	16.5%
	Washing Machine Production (units)	YTD YoY	2.0%	2.0%	7.1%	10.5%	9.3%	10.3%	10.9%	9.5%	9.2%	8.2%	8.1%	6.6%	0.3%	0.3%	3.6%	3.9%	3.5%
	Airconditioner Production (units)	YTD YoY	13.8%	13.8%	14.9%	6.2%	4.7%	4.0%	3.9%	4.3%	3.0%	2.5%	1.2%	0.4%	-2.8%	-2.8%	0.2%	-2.4%	-3.6%
	Freezer Production (units)	YTD YoY	-29.2%	-29.2%	-17.2%	-7.0%	-5.7%	-2.1%	0.5%	2.3%	1.9%	-0.7%	0.0%	0.8%	20.5%	20.5%	14.5%	16.2%	18.4%
	Total Automotive Production (units)	YoY	-34.2%	12.1%	15.3%	8.9%	11.6%	11.4%	13.3%	13.0%	17.1%	12.1%	2.8%	-2.1%	0.01%	-20.5%	-3.0%	-1.7%	-1.2%
	Automotive Sales for Passenger Vehicles (units)	YoY	-32.8%	11.1%	8.2%	11.0%	13.3%	14.5%	14.7%	16.5%	13.2%	7.5%	1.2%	-8.7%	-6.8%	-15.4%	-2.3%	-4.2%	-4.2%
	Automotive Sales for Commercial Vehicles (units)	YoY	-47.6%	29.4%	17.5%	2.7%	-2.0%	9.5%	14.1%	16.3%	29.6%	21.0%	24.4%	15.3%	23.5%	-14.0%	8.9%	8.1%	12.5%
	Automotive Sales for Heavy-Duty Truck (units)	YoY	-28.9%	84.5%	55.1%	37.3%	43.3%	42.3%	83.5%	53.5%	71.5%	55.4%	60.6%	47.9%	47.7%	14.5%	23.3%	16.3%	17.6%
	Automotive Production for Heavy-Duty Truck (units)	YoY	-20%	44%	68%	31%	40%	40%	76%	64%	64%	80%	96%	69%	70%	20%	23%	23%	13%
	Sales of BEV Passenger Vehicles	YoY	-19.4%	33.6%	33.6%	60.5%	45.9%	39.5%	39.7%	37.4%	33.1%	29.8%	23.6%	14.0%	0.4%	-17.6%	-4.2%	-0.8%	24.3%

Source: Bloomberg Finance L.P., NBS, J.P. Morgan Commodities Research

Figure 3: Building new starts growth

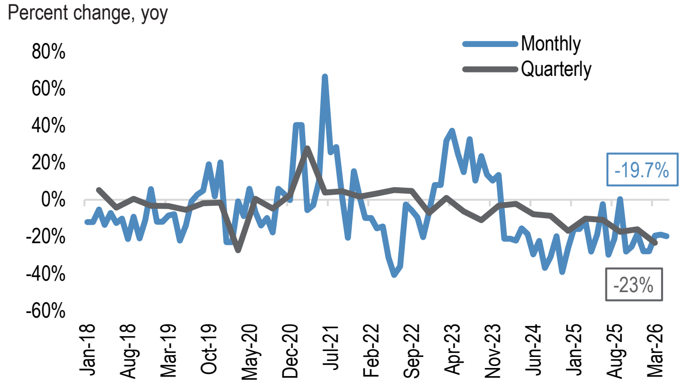
Percent change, yoy



*Includes residential and non-residential. Text box highlights the latest data point; monthly is through May 2026 and quarterly is as of 1Q26.
Source: China NBS

Figure 4: Building completions growth

Percent change, yoy



*Includes residential and non-residential. Text box highlights the latest data point; monthly is through May 2026 and quarterly is as of 1Q26.
Source: China NBS

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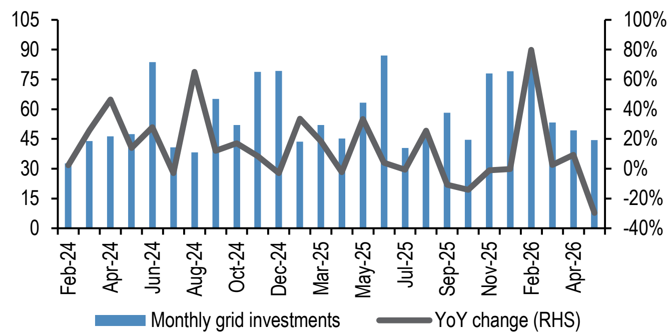
Figure 5: Fixed asset investment in grid infrastructure

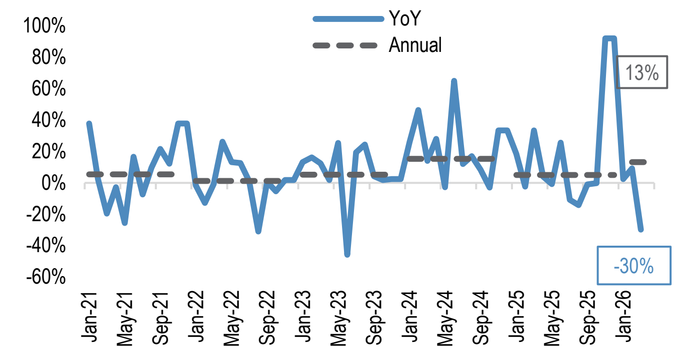
LHS: CNY billions; RHS: Percent change, yoy



*Monthly data through 2025 from NEA. 2026 onwards from Wind. Feb = Jan & Feb combined.
Source: China National Energy Administration, Wind

Figure 6: Fixed asset investment in grid infrastructure

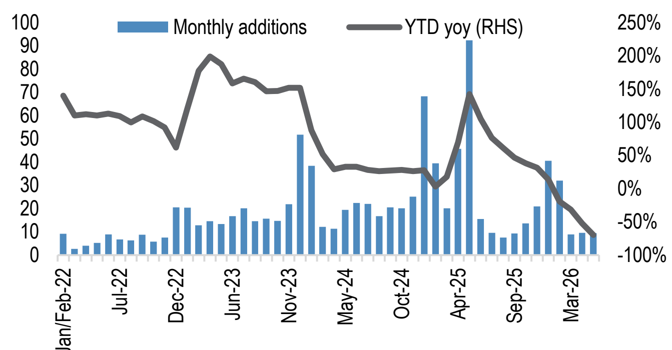
Percent change, yoy



*Monthly data through 2025 from NEA. 2026 onwards from Wind. 2026 is through May.
Source: China National Energy Administration, Wind.

Figure 7: Solar power plant capacity additions

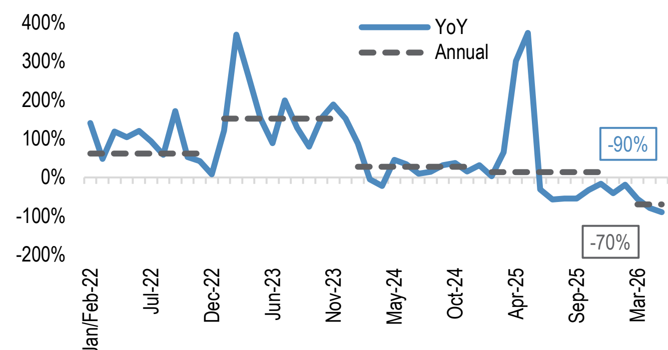
LHS: Gigawatts; RHS: Percent change, YTD yoy



*2026 is through May.
Source: China National Energy Administration

Figure 8: Solar power plant capacity addition growth

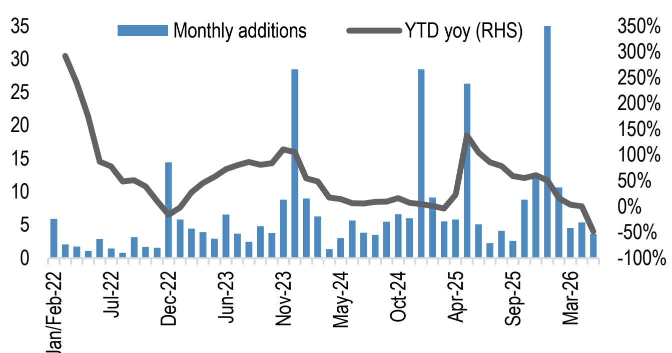
Percent change, yoy



*2026 is through May.
Source: China National Energy Administration

Figure 9: Wind power plant capacity additions

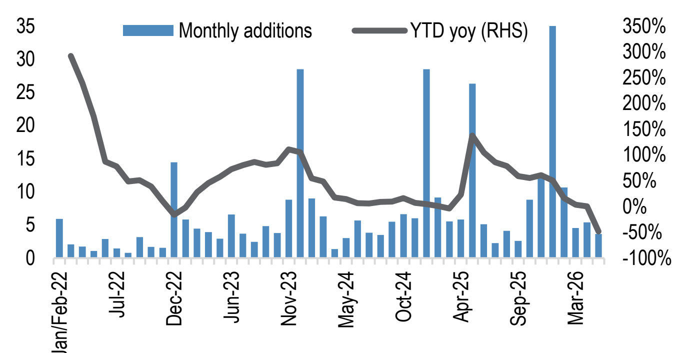
LHS: Gigawatts; RHS: Percent change, yoy



*2026 is through May.
Source: China National Energy Administration

Figure 10: Wind power plant capacity addition growth

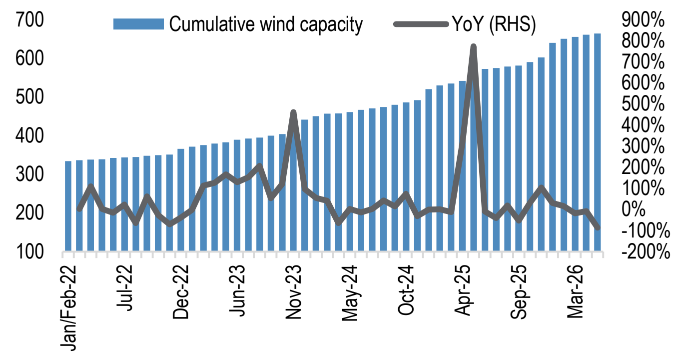
Percent change, yoy



*2026 is through May.
Source: China National Energy Administration

Figure 11: Cumulative wind installed capacity

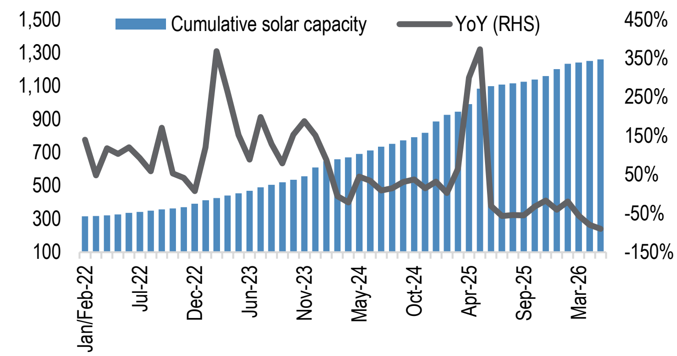
LHS: Gigawatts; RHS: Percent change, yoy of monthly installations



*2026 is through May.
Source: China National Energy Administration

Figure 12: Cumulative solar installed capacity

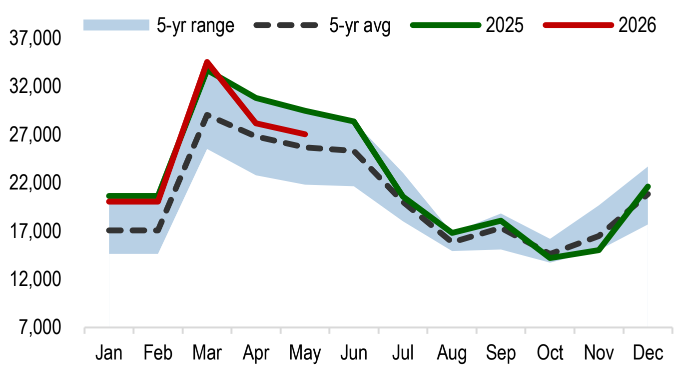
LHS: Gigawatts; RHS: Percent change, yoy of monthly installations



*2026 is through May.
Source: China National Energy Administration

Figure 13: Air conditioner production, monthly

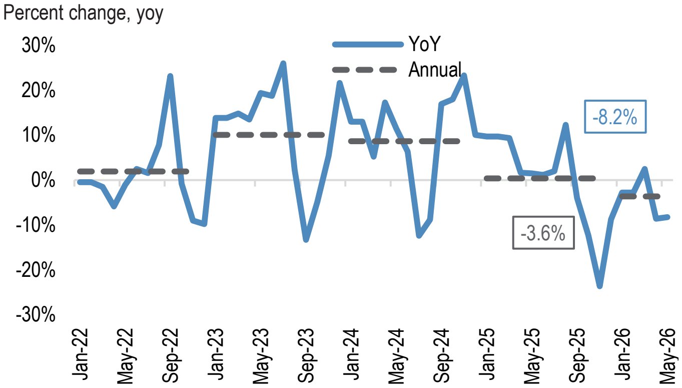
Thousand units



Source: China NBS

Figure 14: Air conditioner production growth

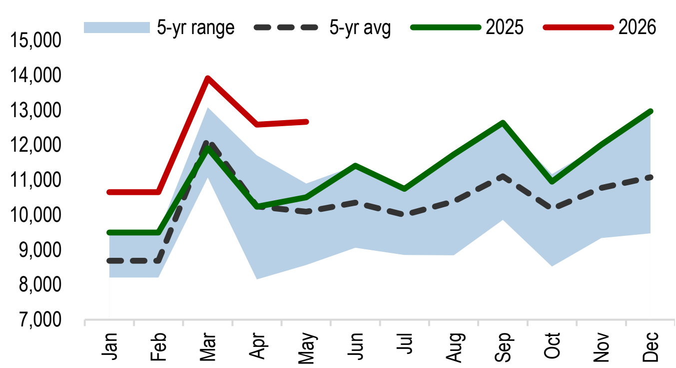
Percent change, yoy



*2026 is through May.
Source: China NBS

Figure 15: Fridge and freezer production, monthly

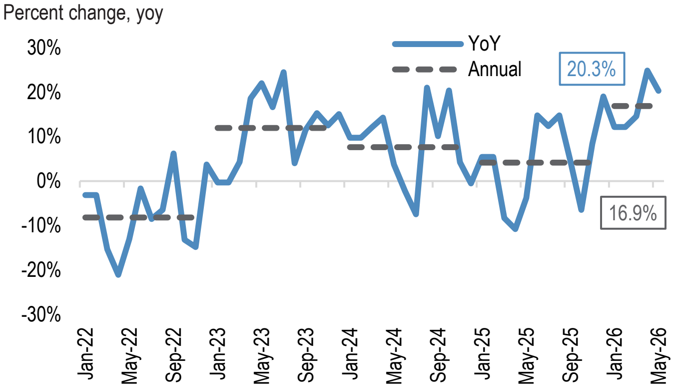
Thousand units



Source: China NBS

Figure 16: Fridge and freezer production growth

Percent change, yoy



*2026 is through May.
Source: China NBS

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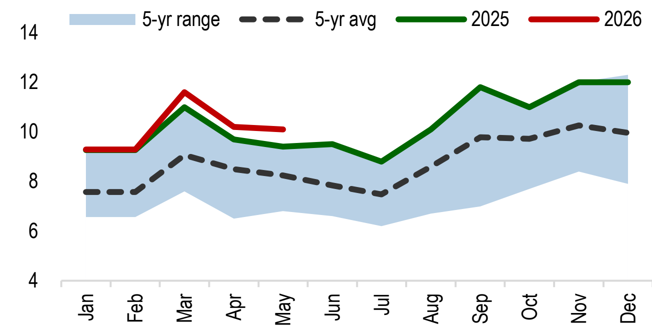
Figure 17: Washing machine production, monthly

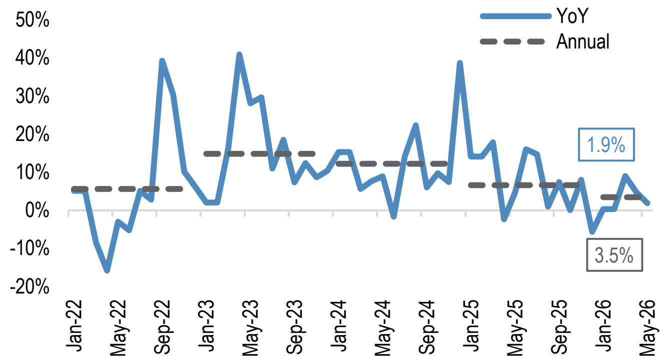
Million units



Source: Source: China NBS

Figure 18: Washing machine production growth

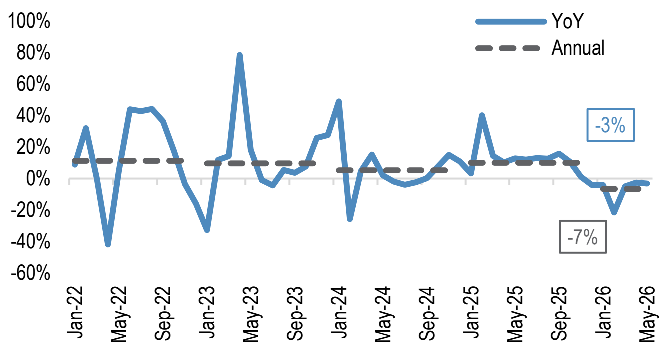
Percent change, yoy



*2026 is through May.
Source: China NBS

Figure 19: Passenger vehicle production growth

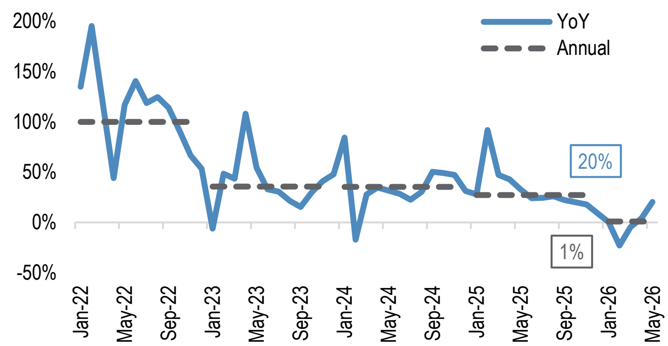
Percent change, yoy



*Includes basic passenger, multi-purpose, SUVs, and crossover vehicles. 2026 is through May.
Source: China Automotive and Research Center

Figure 20: Passenger NEV production growth

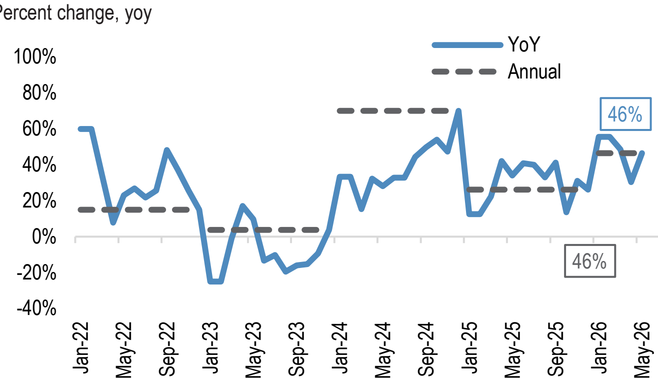
Percent change, yoy



*Includes battery electric vehicles, and plug-in hybrids. 2026 is through May.
Source: China Automotive and Research Center

Figure 21: Industrial robot production growth

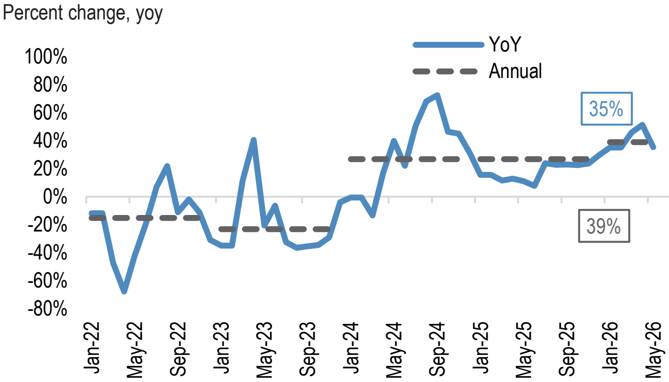
Percent change, yoy



*2026 is through May.
Source: China NBS

Figure 22: Excavator production growth

Percent change, yoy



*2026 is through May.
Source: China NBS

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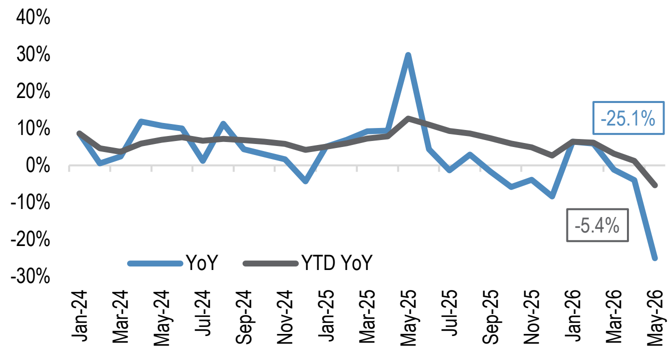
Copper

Figure 23: Consumption-weighted end use indicator for Chinese copper demand growth

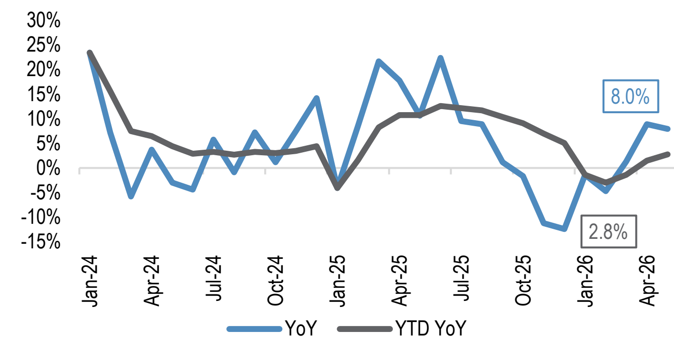
LHS: Percent change, yoy; RHS: Percent change, YTD yoy



*This indicator applies consumption-weights to underlying reported end-use production/installation/export growth figures to derive an estimate for monthly aggregate Chinese copper demand growth. It covers construction, grid, solar, wind, EVs, other transport, consumer durables, ACs/refrigeration, machinery and product and cable exports which, combined, roughly account for 90% of Chinese end use demand. Data through May 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 24: China copper apparent demand growth

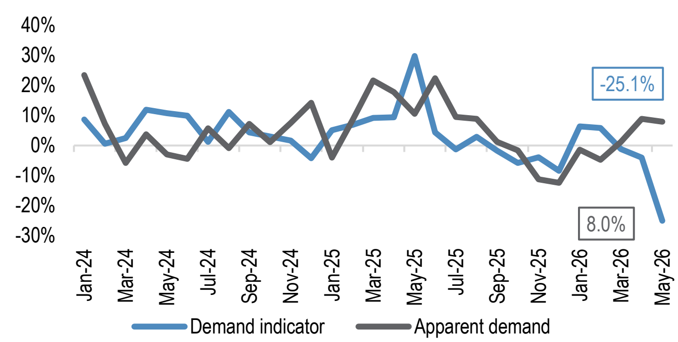
Percent change



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE. Data through May 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 25: YoY Growth – Consumption-weighted end use copper demand indicator vs apparent demand

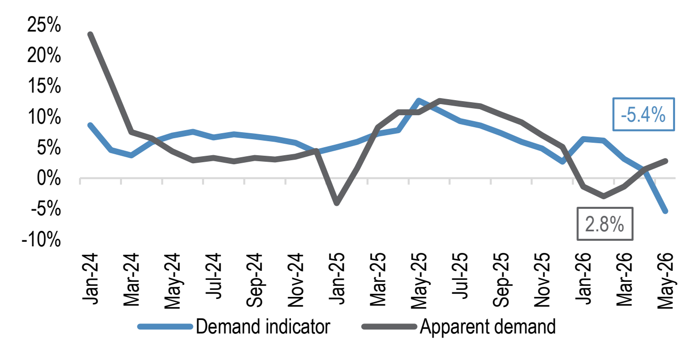
Percent change, yoy



*See explanatory notes in Figures 24 & 25. Data through May 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 26: YTD YoY Growth – Consumption-weighted end use copper demand indicator vs apparent demand

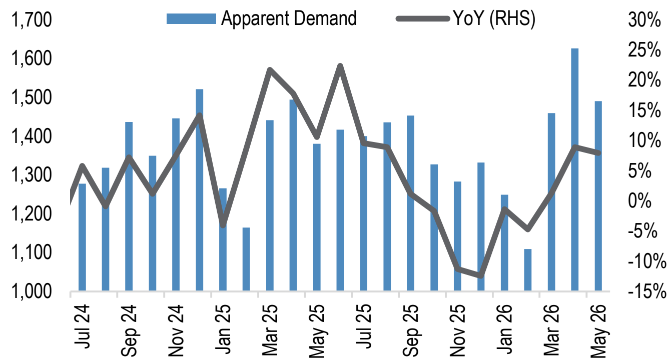
Percent change, YTD yoy



*See explanatory notes in Figures 24 & 25. Data through April 2026.
Source: China National Energy Administration, China Automotive Tech and Research Center, Tianjin, NBS, China Customs, Bloomberg Finance L.P., J.P. Morgan Commodities Research

Figure 27: China apparent copper demand

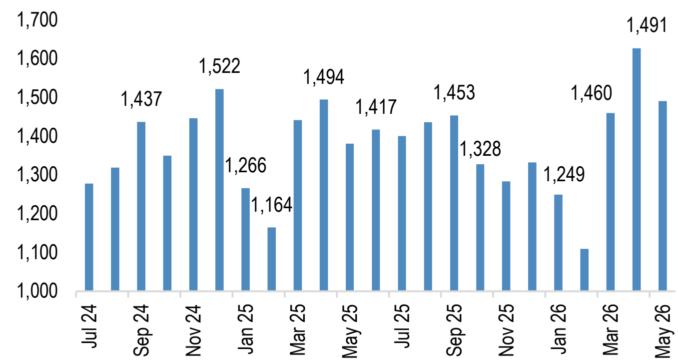
LHS: Thousand mt; RHS: Percent change, yoy



Source: China Customs General Administration, SHFE. *Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.

Figure 28: China apparent copper demand

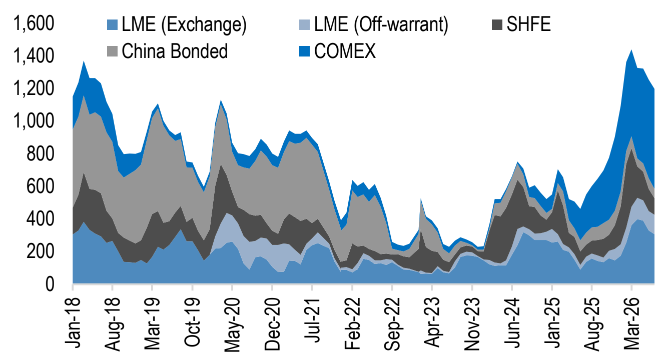
Thousand mt



Source: China Customs General Administration, SHFE. *Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.

Figure 29: Global visible copper inventories

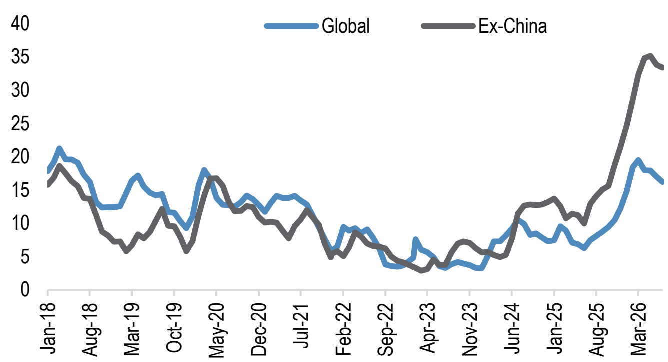
Thousand mt



Source: CRU, LME, SHFE, COMEX. Bonded includes Chinese bonded stocks in Shanghai, Guangdong and Qingdao. As of 4th June 2026, while LME off-warrant as of 10th July 2026.

Figure 30: Global visible copper inventory in days of use

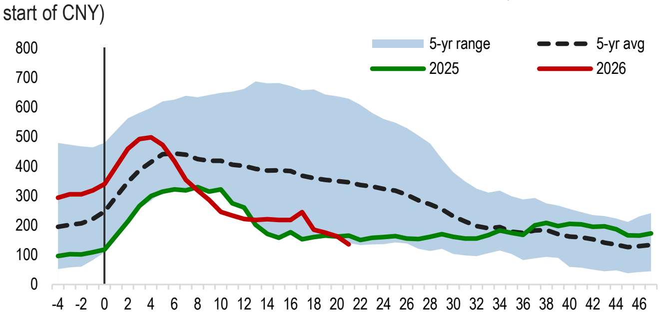
Days



Source: SHFE, CRU, SMM, J.P. Morgan Commodities Research

Figure 31: Total visible China copper inventory (SHFE + Bonded)

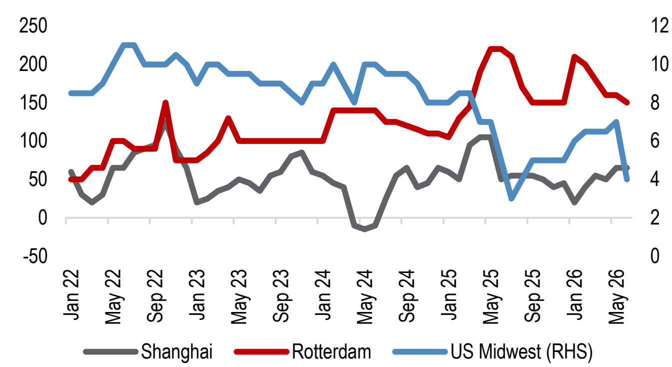
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, CRU, SMM, J.P. Morgan Commodities Research

Figure 32: Copper regional premiums

LHS: US$/t; RHS: US$c/lb



Source: CRU

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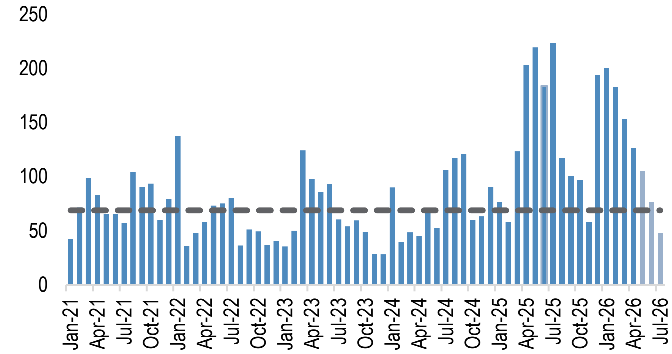
Figure 33: Monthly US refined copper imports

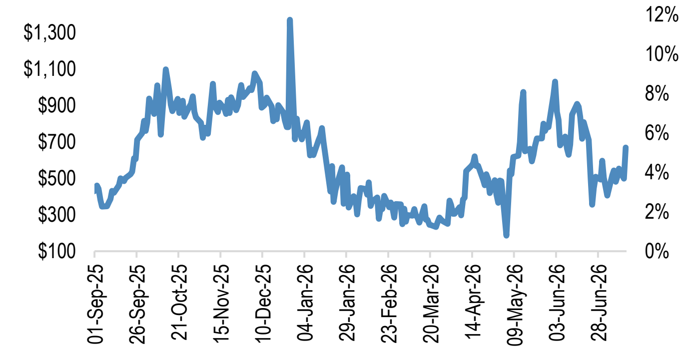
LHS: Thousand mt, RHS: US$/mt



Note: 1st May- 9th July 2026 estimated based on bill of lading data.
Source: US Census Bureau, IHS Markit, COMEX, LME, J.P. Morgan Commodities Research

Figure 34: Dec'26 COMEX/LME copper spread

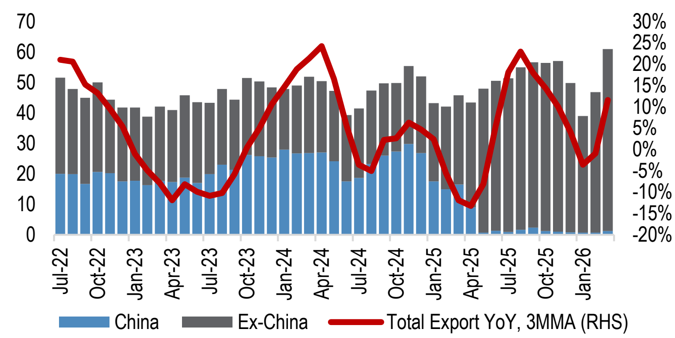
LHS: US$/mt, RHS: Percent of LME price (implied tariff rate)



Source: Bloomberg Finance L.P. *Arb as of London, 14th July 2026.

Figure 35: Monthly US copper scrap exports

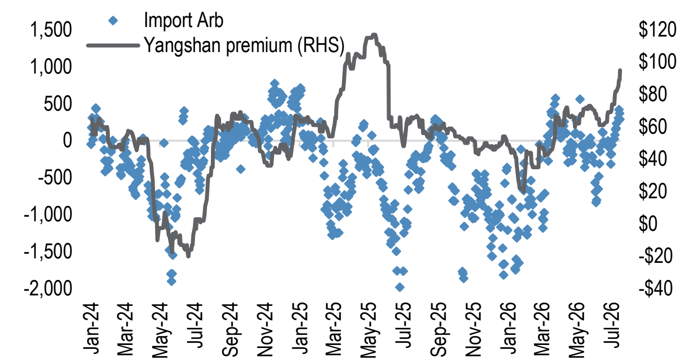
LHS: Thousand mt (cu contained); RHS: Percent change, YoY



*Data as of March 2026.
Source: US Census Bureau, J.P. Morgan Commodities Research

Figure 36: Chinese spot copper import arbitrage and onshore spot premium/discount

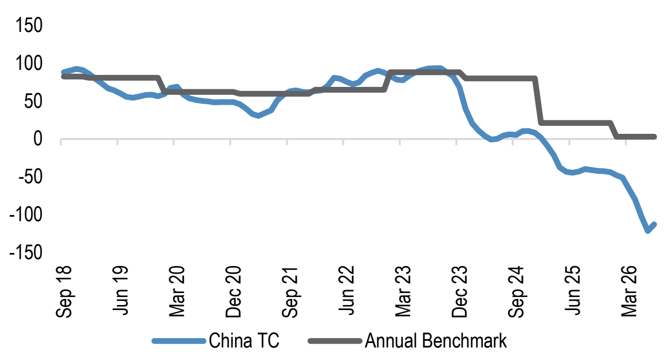
LHS: CNY/mt ; RHS: US$/mt



Source: SMM, Bloomberg Finance L.P., J.P. Morgan Commodities Research. *Arb as of 8:00 AM London. As of 9th July 2026.

Figure 37: China copper imported concentrate treatment charges vs annual benchmark

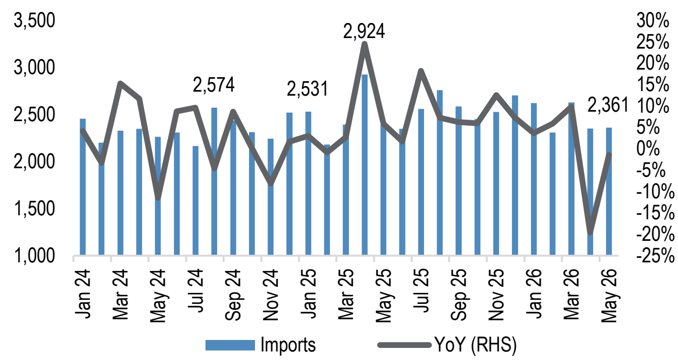
US$/mt



Source: Metal Bulletin, Platts, Wood Mackenzie

Figure 38: China copper concentrate imports

LHS: Thousand mt (gross); RHS: Percent change, yoy



Source: China Customs General Administration

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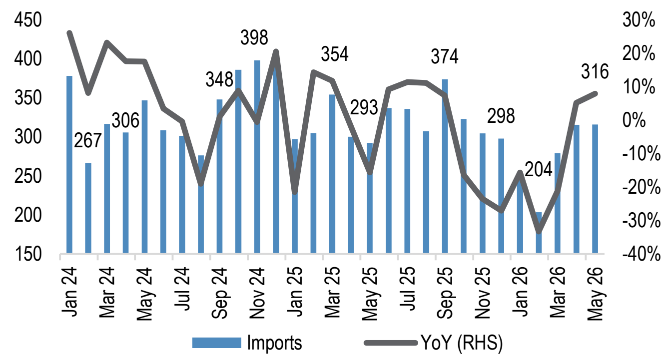
Figure 39: China refined copper imports

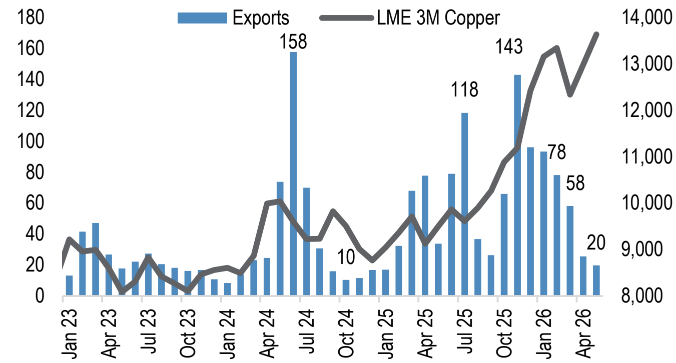
LHS: Thousand mt; RHS: Percent change, yoy



Source: China Customs General Administration

Figure 40: China refined copper exports vs LME 3-month copper price

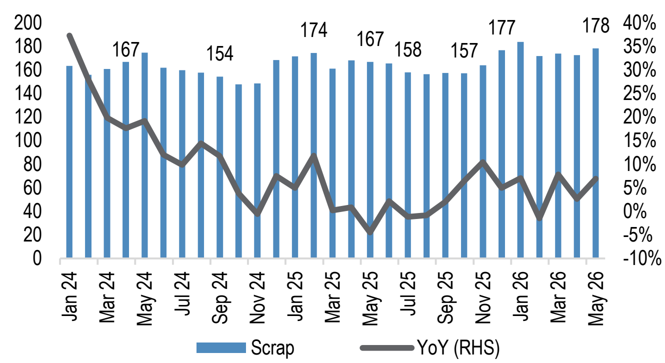
LHS: Thousand mt; RHS: US$/mt



Source: China Customs General Administration, Bloomberg Finance L.P.

Figure 41: China contained copper scrap imports, 3mma

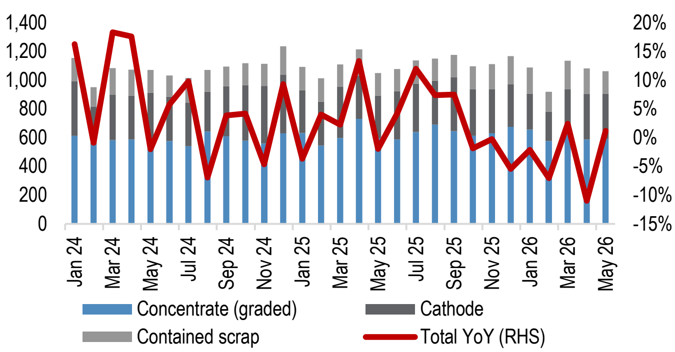
LHS: Thousand mt (Cu contained); RHS: Percent change, yoy



Source: China Customs General Administration

Figure 42: China copper imports of concentrate, cathode and contained scrap

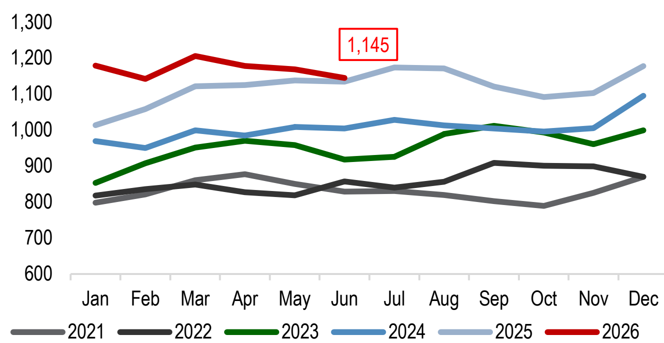
LHS: Thousand mt; RHS: Percent change, yoy



*Assumes 25% copper contained in concentrate.
Source: China Customs General Administration

Figure 43: China refined copper production, monthly

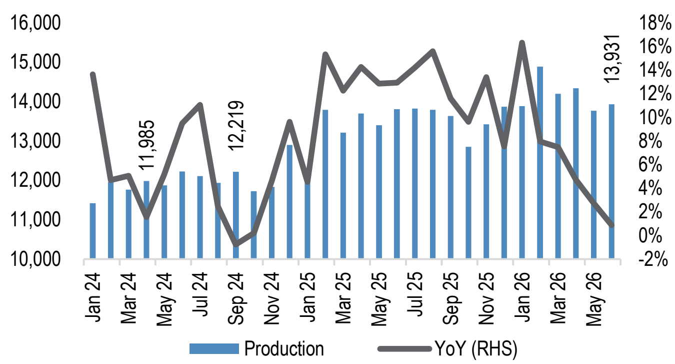
Thousand mt



Source: SMM

Figure 44: Annualized China refined copper production

LHS: Thousand mt; RHS: Percent change, yoy



Source: SMM

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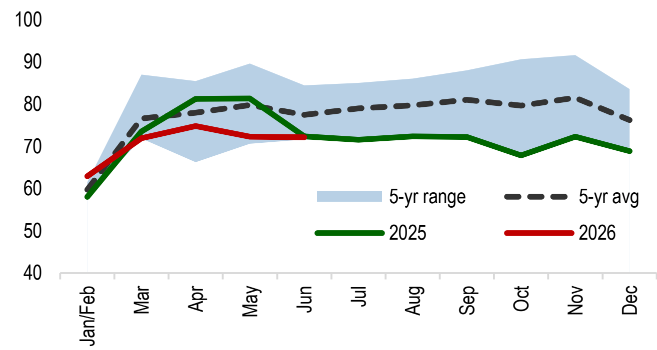
Figure 45: China copper wire and cable producer operating rates

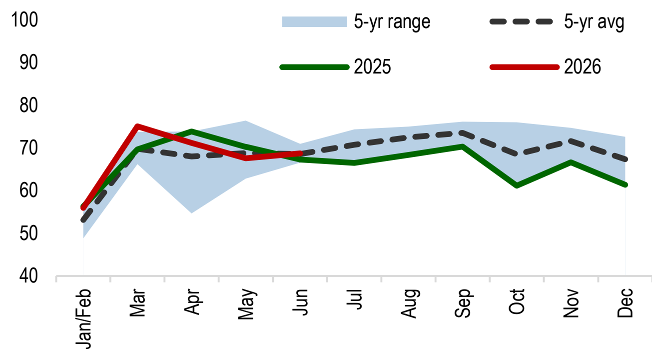
Percent



Source: SMM

Figure 46: China copper cathode-fed rod producer operating rates

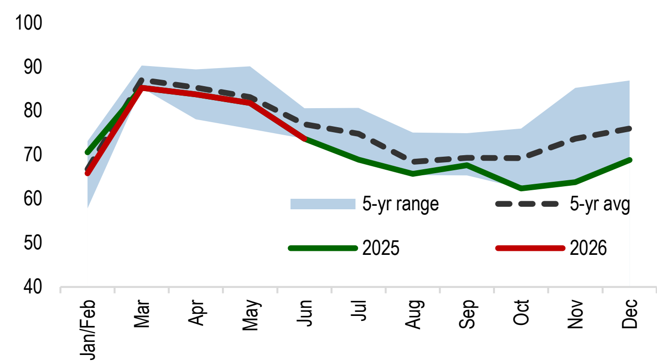
Percent



Source: SMM

Figure 47: China copper tube producer operating rates

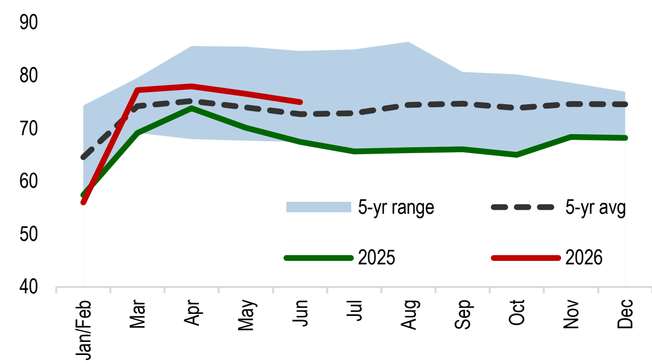
Percent



Source: SMM

Figure 48: China copper flat-rolled producer operating rates

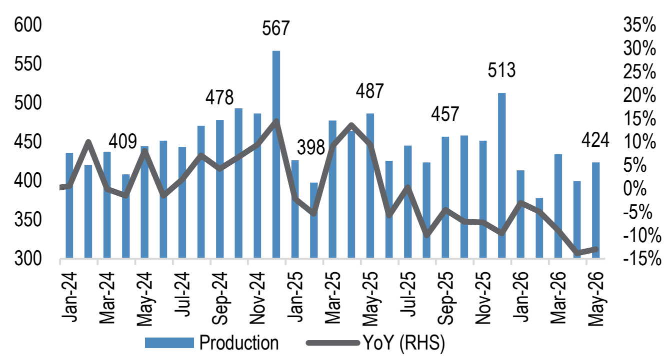
Percent



Source: SMM

Figure 49: Chile copper mine production

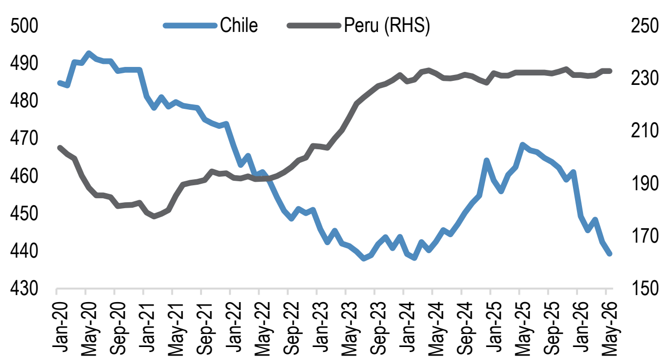
LHS: Thousand metric tonnes; RHS: Percent change yoy



Source: Instituto National de Estadistico

Figure 50: Chile and Peru copper mine production, 12mma

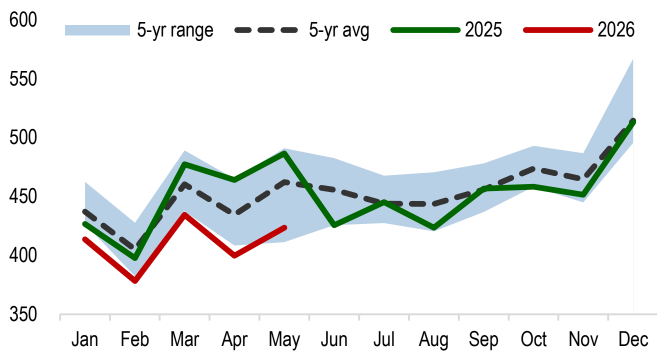
Thousand metric tonnes



Source: Instituto National de Estadistico, Ministerio de Energias y Minas

Figure 51: Chile copper mine production, monthly

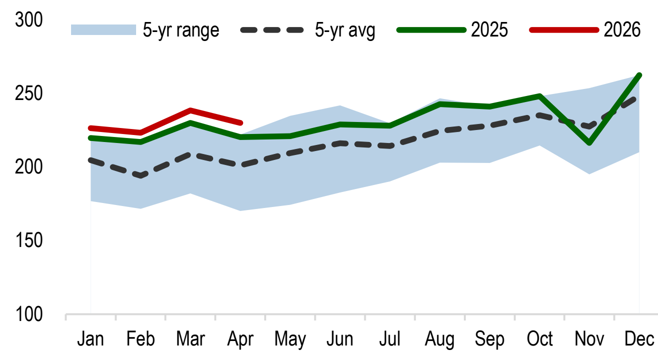
Thousand mt



Source: Instituto National de Estadistico

Figure 52: Peru copper mine production, monthly

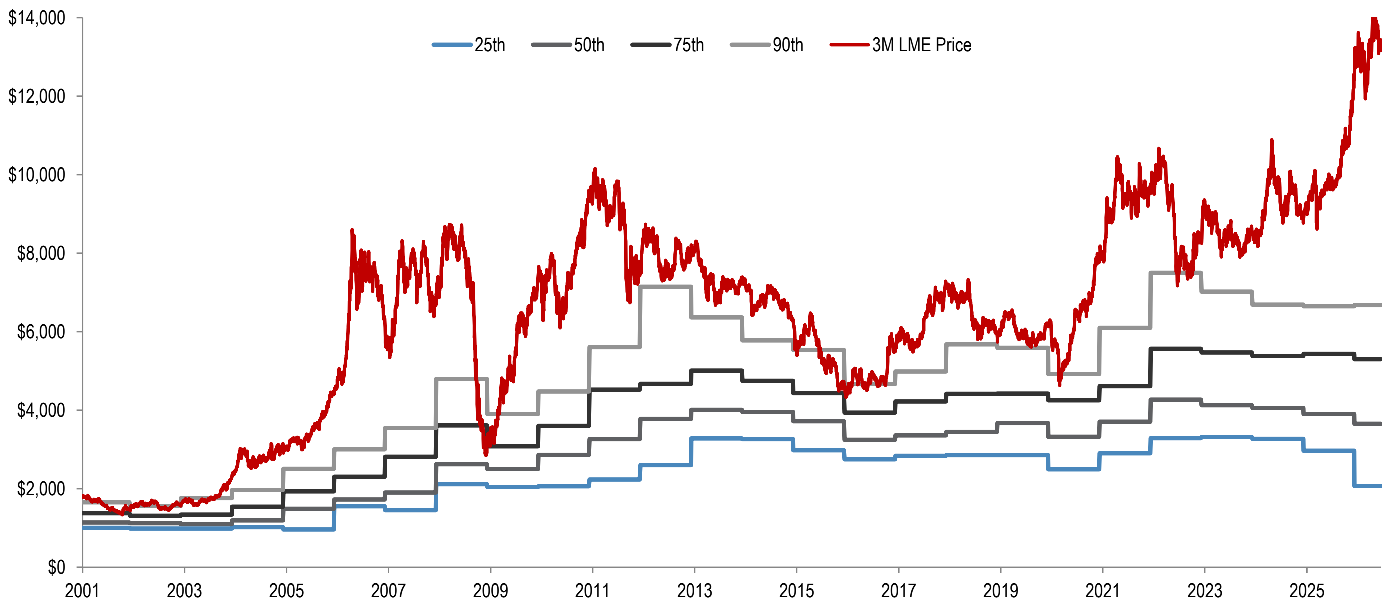
Thousand mt



Source: Ministerio de Energias y Minas

Figure 53: Global copper mining C1 + sustaining capex costs by %-tile and LME 3M copper price

US$/mt MOD



Source: Wood Mackenzie, J.P. Morgan Commodities Research. *LME prices as of 9th July 2026.

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Table 5: China copper trade flow

Thousand mt; Percent Change, yoy.

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Copper Concentrate Imports (Gross)	19,720	22,024	21,777	23,428	25,318	27,588	28,165	30,365	12,276
Refined Copper Imports	3,753	3,550	4,670	3,627	3,885	3,739	4,042	3,830	1,366
Unwrought Copper and Products Imports	5,293	4,991	6,686	5,530	5,872	5,504	5,700	5,350	2,020
Contained Copper Imports (from Scrap)	1,434	1,239	733	1,360	1,588	1,721	1,937	1,983	857
Refined Copper Exports	282	316	190	266	253	279	458	795	275
Unwrought Copper & Copper Products Exports	793	841	676	933	917	958	1,287	1,673	724
Net Refined Imports	3,472	3,234	4,480	3,361	3,632	3,459	3,584	3,035	1,090
YoY-%									
Copper Concentrate Imports		11.7%	-1.1%	7.6%	8.1%	9.0%	2.1%	7.8%	-1.2%
Refined Copper Imports		-5.4%	31.5%	-22.3%	7.1%	-3.8%	8.1%	-5.3%	-11.9%
Unwrought Copper and Products Imports		-5.7%	34.0%	-17.3%	6.2%	-6.3%	3.6%	-6.1%	-7.3%
Contained Copper Imports (from scrap)		-13.6%	-40.8%	85.4%	16.8%	8.4%	12.5%	2.4%	3.6%
Refined Copper Exports		12.4%	-40.1%	40.5%	-5.2%	10.6%	63.9%	73.6%	20.2%
Unwrought Copper & Copper Products Exports		6.1%	-19.6%	38.0%	-1.8%	4.5%	34.3%	30.0%	21.8%
Net Refined Imports		-6.8%	38.5%	-25.0%	8.1%	-4.8%	3.6%	-15.3%	-17.4%

*Assumes 25% copper contained in concentrate. 2026 is through May.
Source: China Customs General Administration

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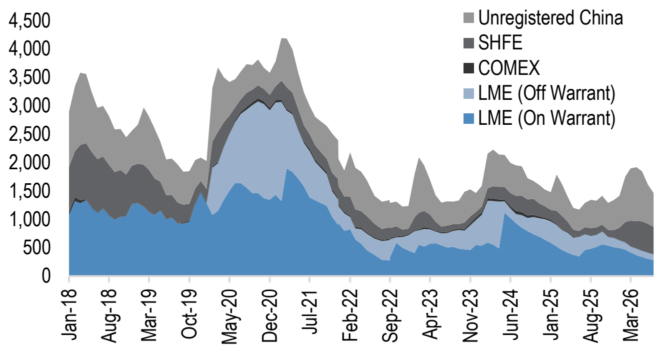
Aluminium

Figure 54: Global visible aluminium inventories

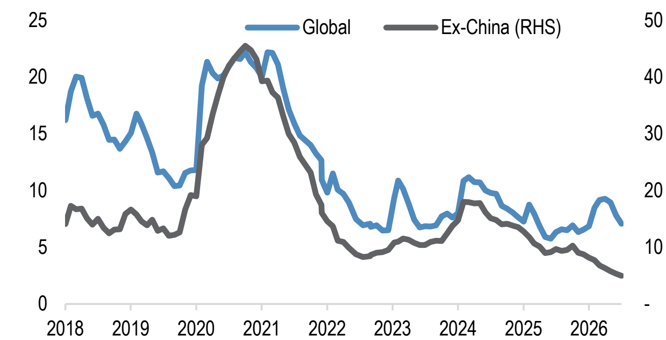
Thousand mt



Source: CRU, LME, SHFE, COMEX. Regional includes stocks in Shanghai, Jiangsu, Nanhai, Zhejiang, Tianjin, Henan, Shandong, Chongqing. As of 14th July 2026.

Figure 55: Global visible aluminium inventory in days of use

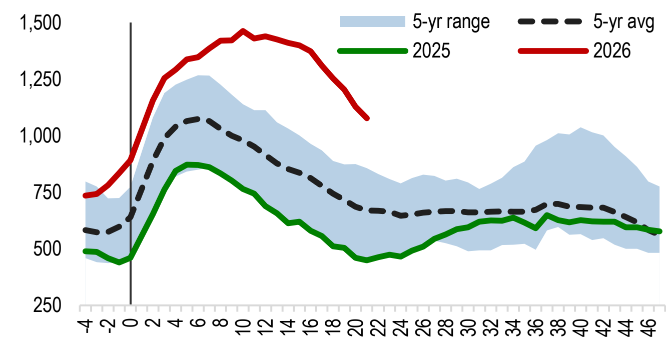
Days



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 56: Total visible China aluminium inventory (SHFE + regional warehouses)

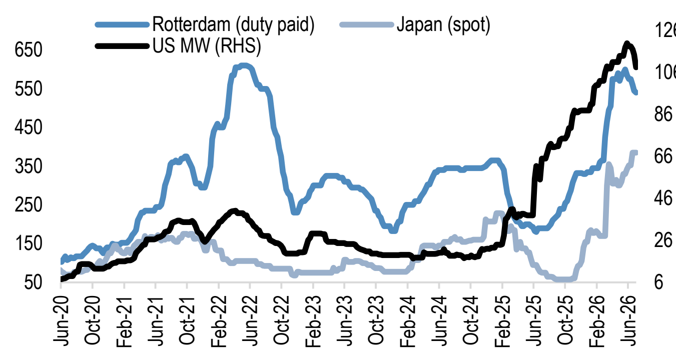
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, CRU, SMM, J.P. Morgan Commodities Research

Figure 57: Regional aluminium ingot premiums

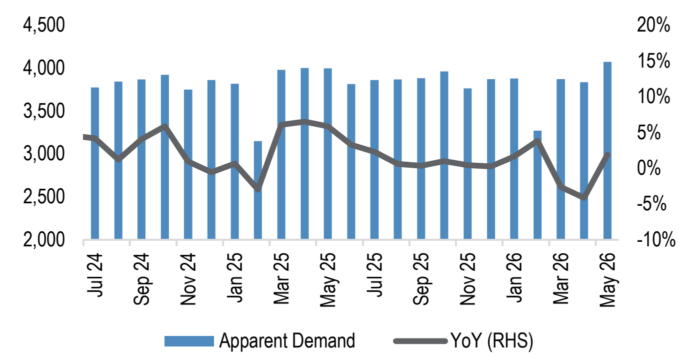
LHS: US$/mt; RHS: c/lb



Source: CRU

Figure 58: China apparent aluminium demand

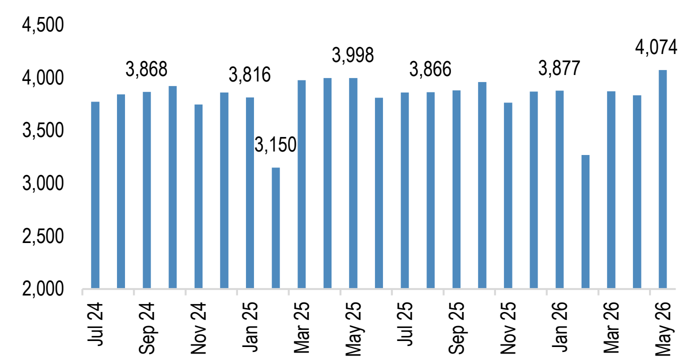
LHS: Thousand mt; RHS: Percent change, yoy



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.
Source: CRU, China Customs General Administration, SHFE

Figure 59: China apparent aluminium demand

Thousand metric tonnes



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.
Source: CRU, China Customs General Administration, SHFE

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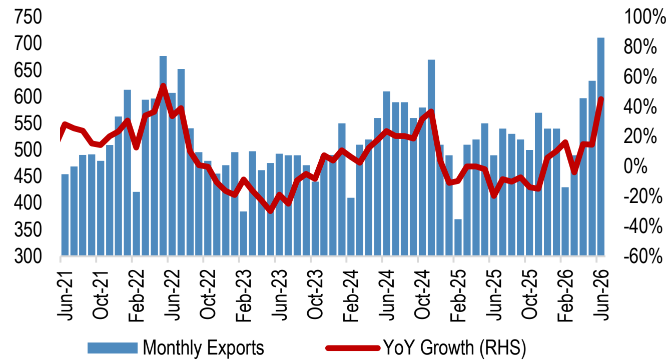
Figure 60: Chinese unwrought aluminium and aluminium product exports

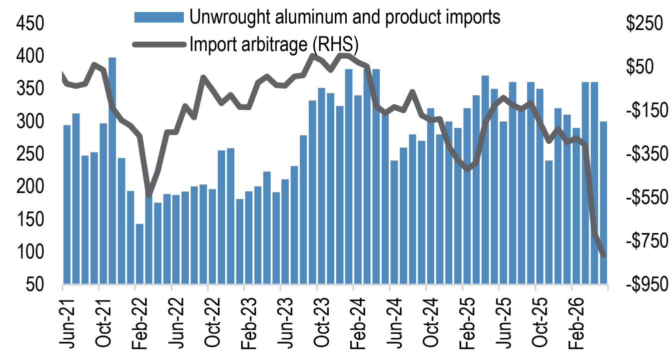
LHS: Thousand mt; RHS: Percent change, yoy



Source: China Customs, J.P. Morgan Commodities Research

Figure 61: Chinese unwrought aluminium and aluminium product imports

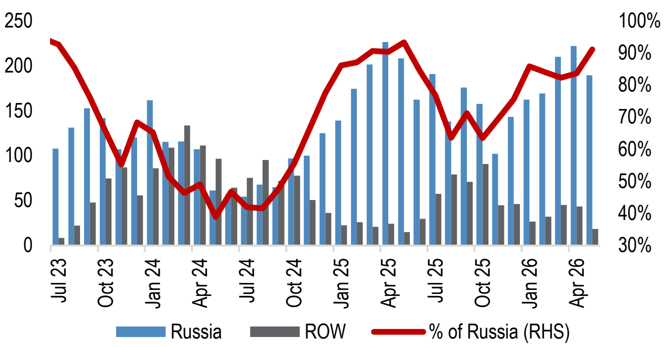
LHS: Thousand mt; RHS: US$/mt



Source: China Customs, J.P. Morgan Commodities Research

Figure 62: China refined aluminium imports by region

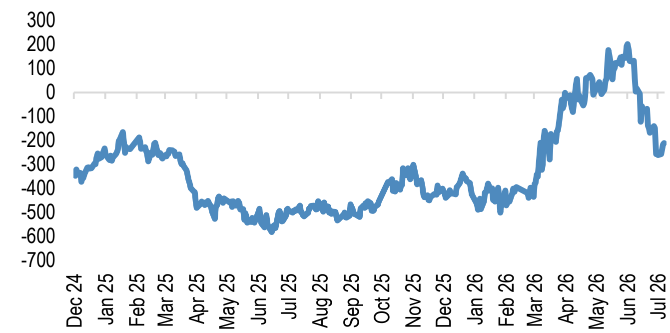
LHS: Thousand mt; RHS: Percent of total



Source: China Customs, J.P. Morgan Commodities Research

Figure 63: China aluminium semis export arb proxy (China to rest of Asia)

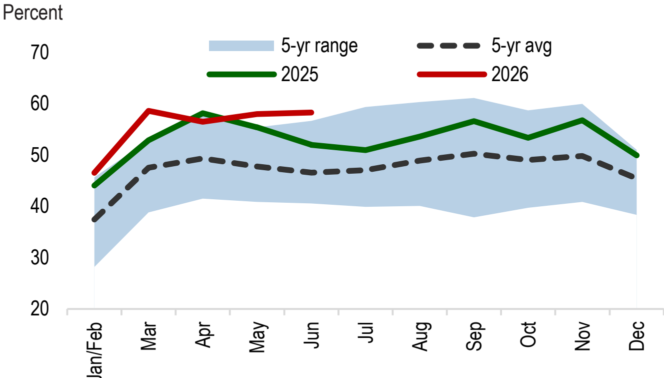
US$/mt. Differential between China spot prices (inclusive of VAT) and LME + Asia premium



Note: Export arb is calculated as the differential between China spot prices and LME + Asia premium less conversion costs (ingot to semis), logistics, shipping and financing costs.
Source: SHFE, SMM, Shanghai Metals Exchange, LME, Bloomberg L.P., J.P. Morgan Commodities Research. *Prices as of 9th July 2026.

Figure 64: China aluminium wire & rod producer operating rates

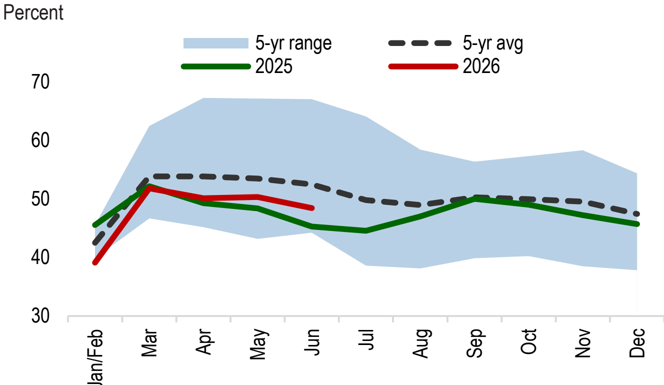
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 65: China aluminium profile producer operating rates

Percent



Source: SMM, Bloomberg Finance L.P.

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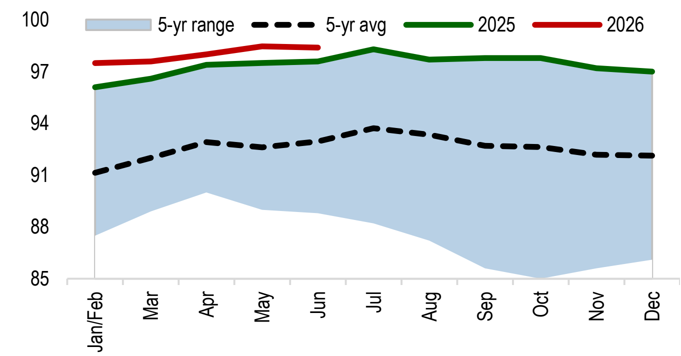
Figure 66: China aluminium ingot producer operating rates

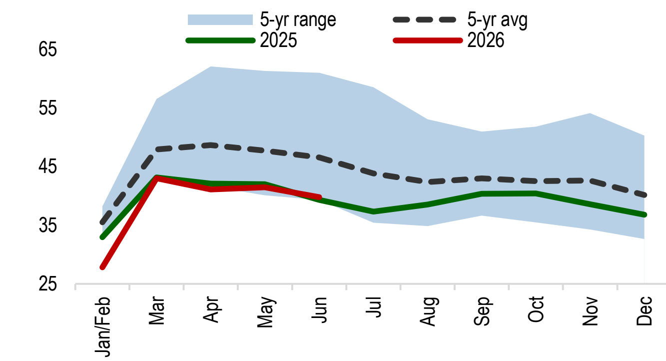
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 67: China aluminium extrusions producer operating rates

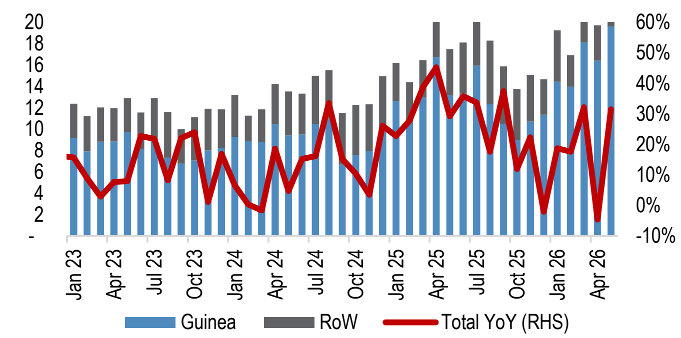
Percent



Source: SMM, Bloomberg Finance L.P.

Figure 68: China total bauxite imports

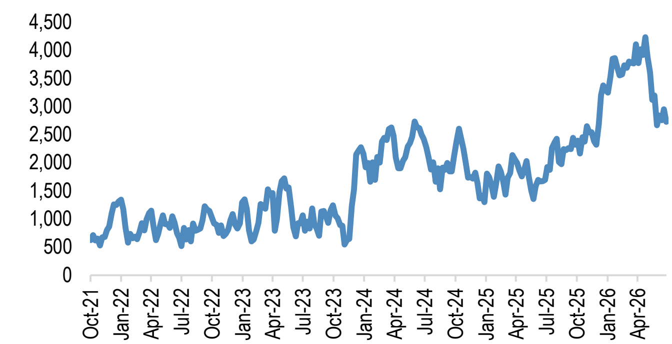
LHS: Million mt; RHS: Percent change yoy



Source: China Customs

Figure 69: China imports of bauxite from Guinea, 3-week ma

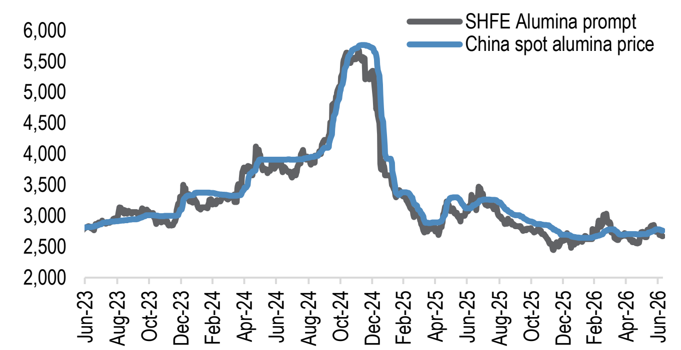
Thousand mt



Source: Mysteel Global

Figure 70: China alumina price

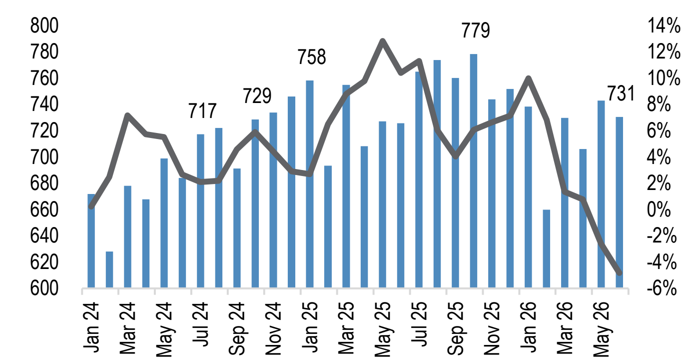
CNY/mt



Source: SHFE, Asian Metal Inc, Bloomberg Finance L.P.

Figure 71: China monthly alumina production

LHS: Thousand mt; RHS: Percent change



Source: SMM, Bloomberg Finance L.P.

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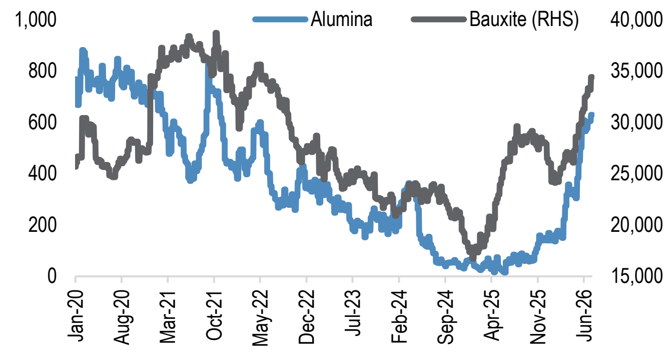
Figure 72: China port inventory of bauxite and alumina

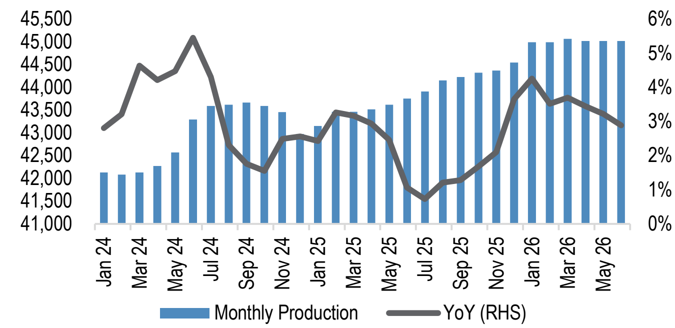
Thousand mt



Source: Mysteel Global

Figure 73: Annualized Chinese refined monthly aluminium production

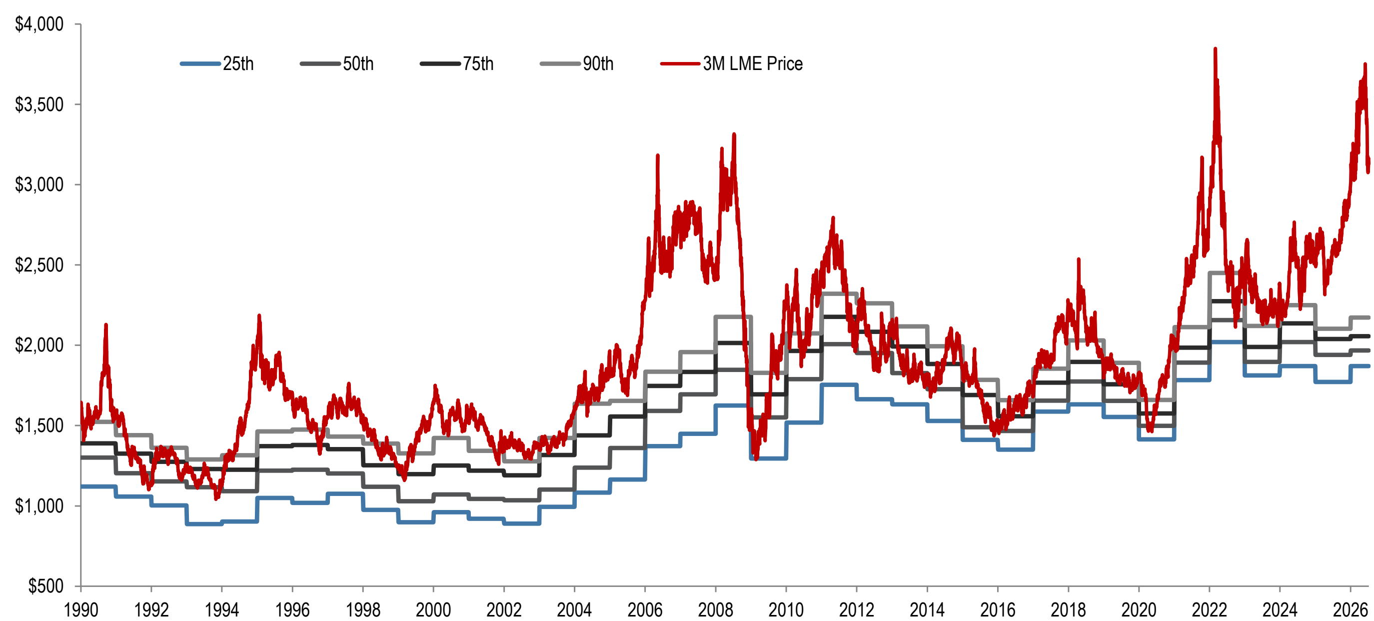
LHS: Thousand mt; RHS: Percent change



Source: CRU

Figure 74: Global primary aluminium smelting C1 costs by %-tile and LME 3M aluminium price

US$/mt MOD



Source: Wood Mackenzie, LME, J.P. Morgan Commodities Research. *LME prices as of 4th July 2026.

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Table 6: China aluminium trade flow

Thousand mt; Percent Change, yoy.

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Bauxite Imports	82,689	100,665	111,537	107,421	125,689	141,647	159,179	201,185	100,758
Refined Aluminium Imports	125	75	1,064	1,580	668	1,543	2,136	2,547	1,119
Unwrought Aluminium and Products Imports	596	646	2,704	3,214	2,391	3,060	3,740	3,920	1,620
Aluminum Waste and Scrap Imports	1,565	1,395	825	1,033	1,518	1,753	1,785	2,016	152
Refined Aluminium Exports	55	76	10	7	196	150	121	298	22
Unwrought Aluminium & Aluminium Products Exports	5,800	5,735	4,860	5,620	6,604	5,679	6,660	6,130	630
Net Refined Imports	70	-1	1,055	1,573	472	1,393	2,015	2,249	1,097
YoY-%									
Bauxite Imports		21.7%	10.8%	-3.7%	17.0%	12.7%	12.4%	26.4%	18.1%
Refined Aluminium Imports		-39.5%	1309.8%	48.4%	-57.7%	131.0%	38.4%	19.2%	5.8%
Unwrought Aluminium and Products Imports		8.3%	318.8%	18.8%	-25.6%	28.0%	22.2%	4.8%	-3.0%
Aluminum Waste and Scrap Imports		-10.8%	-40.9%	25.3%	46.9%	15.5%	1.8%	12.9%	-82.3%
Refined Aluminium Exports		38.0%	-87.3%	-27.1%	2680.0%	-23.4%	-19.4%	145.7%	-97.4%
Unwrought Aluminium & Aluminium Products Exports		-1.1%	-15.3%	15.6%	17.5%	-14.0%	17.3%	-8.0%	-74.2%

*2026 is through May.
Source: China Customs General Administration

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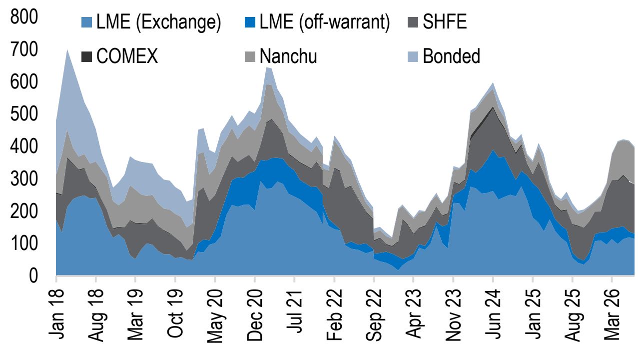
Zinc

Figure 75: Global visible zinc inventories

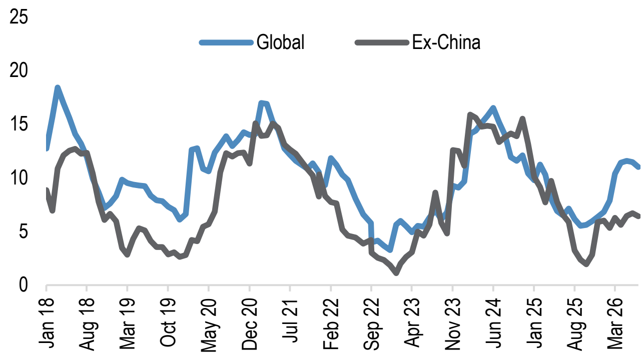
Thousand metric tonnes



Source: CRU, LME, SHFE, COMEX. Nanchu and Bonded warehouses are in China. As of 10th July 2026.

Figure 76: Global visible zinc inventories in days of use

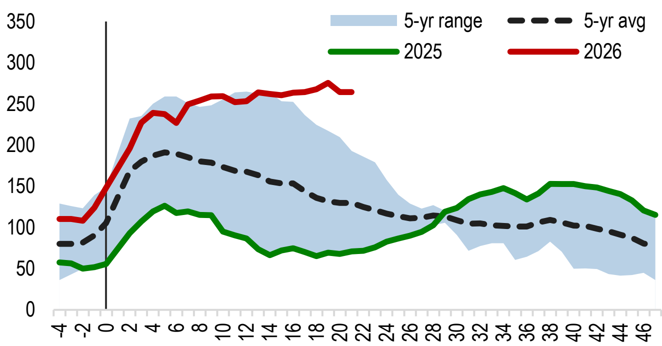
Days



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 77: Total visible China zinc inventory (SHFE + Regional Warehouses)

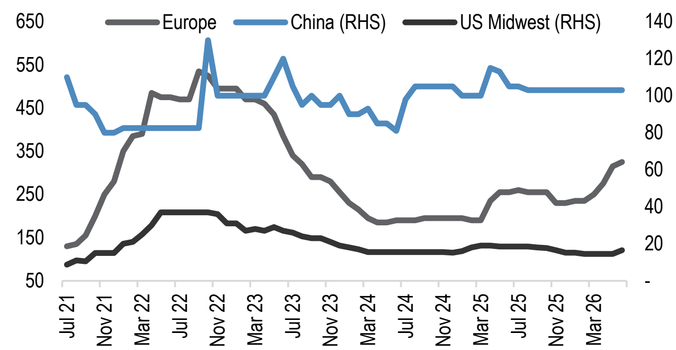
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 78: Regional zinc premiums

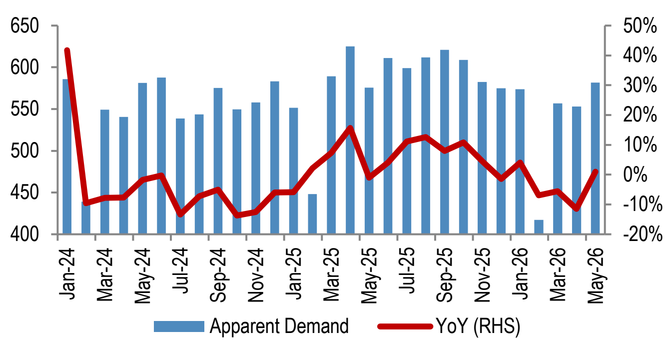
US$/mt



Source: CRU

Figure 79: China apparent zinc demand

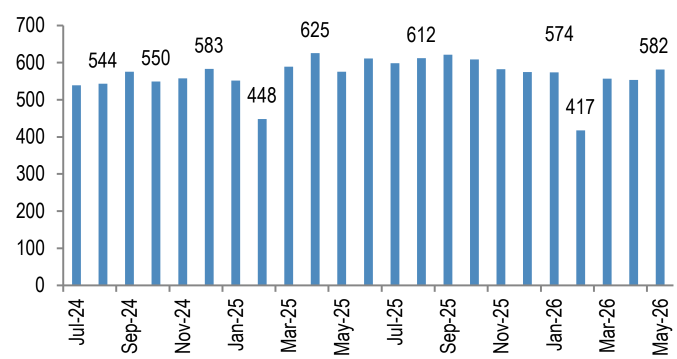
LHS: Thousand mt; RHS: Percent change, yoy



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE
Source: China Customs General Administration, SHFE

Figure 80: China apparent zinc demand

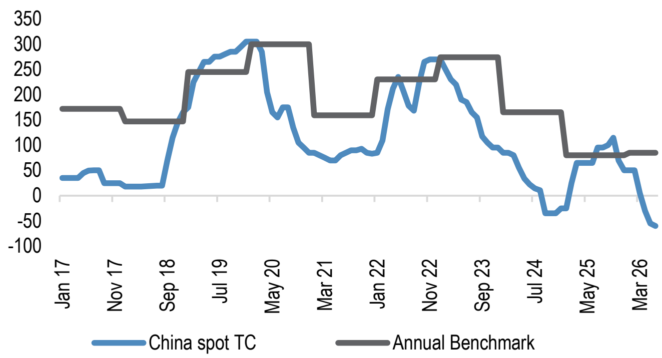
Thousand mt



*Apparent demand is calculated as (Imports-Exports)+Production-Monthly stock changes on the SHFE.
Source: China Customs General Administration, SHFE

Figure 81: China zinc spot concentrate treatment charges vs annual benchmark

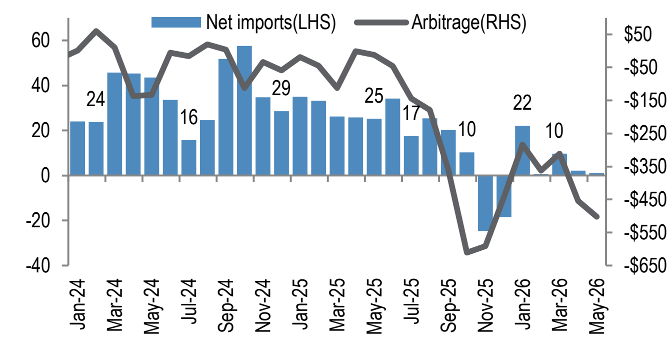
US$/mt



Source: Metal Bulletin, Platts, Wood Mackenzie

Figure 82: China net imports of refined zinc and China's zinc import arb

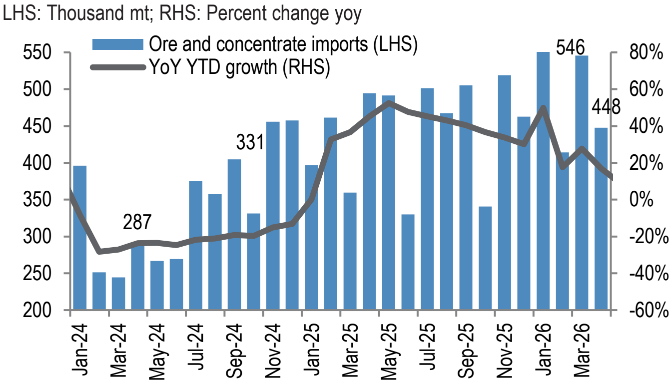
LHS: Thousand mt; RHS: US$/mt



Source: China Customs, CRU, J.P. Morgan Commodities Research

Figure 83: China zinc concentrate imports

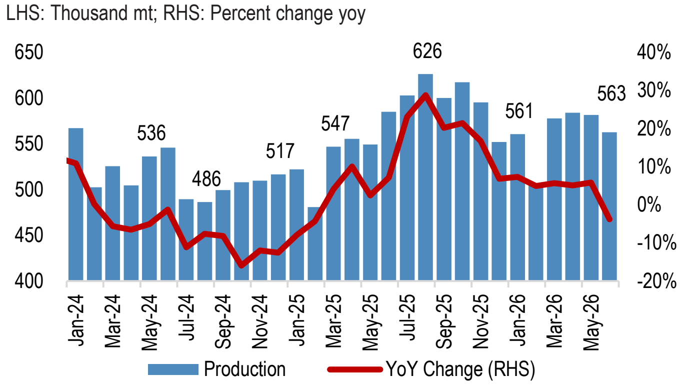
LHS: Thousand mt; RHS: Percent change yoy



Source: NBS, Bloomberg Finance L.P.

Figure 84: China monthly refined zinc production

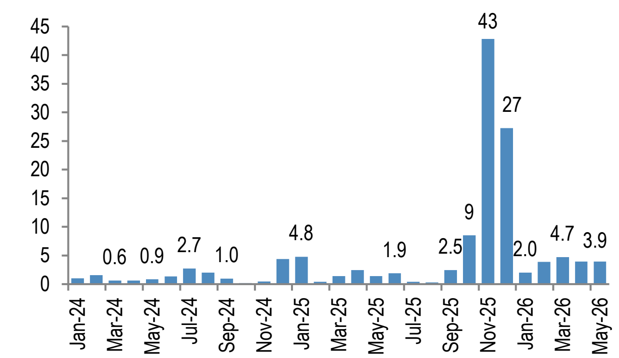
LHS: Thousand mt; RHS: Percent change yoy



Source: SMM, Bloomberg Finance L.P.

Figure 85: China refined zinc exports

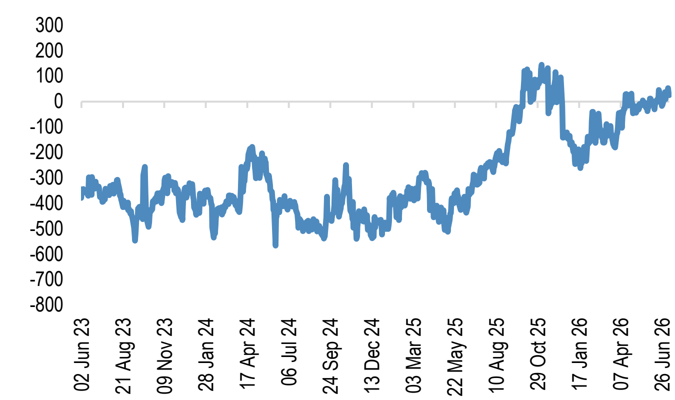
Thousand mt



Source: NBS, Bloomberg Finance L.P.

Figure 86: China zinc export arb proxy

US$/mt. Differential between China spot prices (inclusive of VAT) and LME 3M + Asia premium



Source: SHFE, SMM, Shanghai Metals Exchange, LME, Bloomberg L.P., J.P. Morgan Commodities Research. *Prices as of 9th July 2026.

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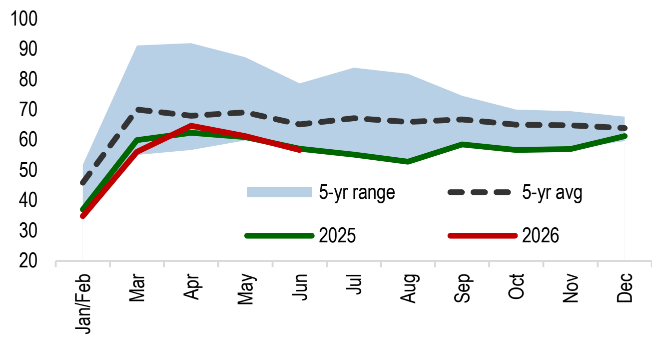
Figure 87: China zinc galvanizers operating rates

Percent

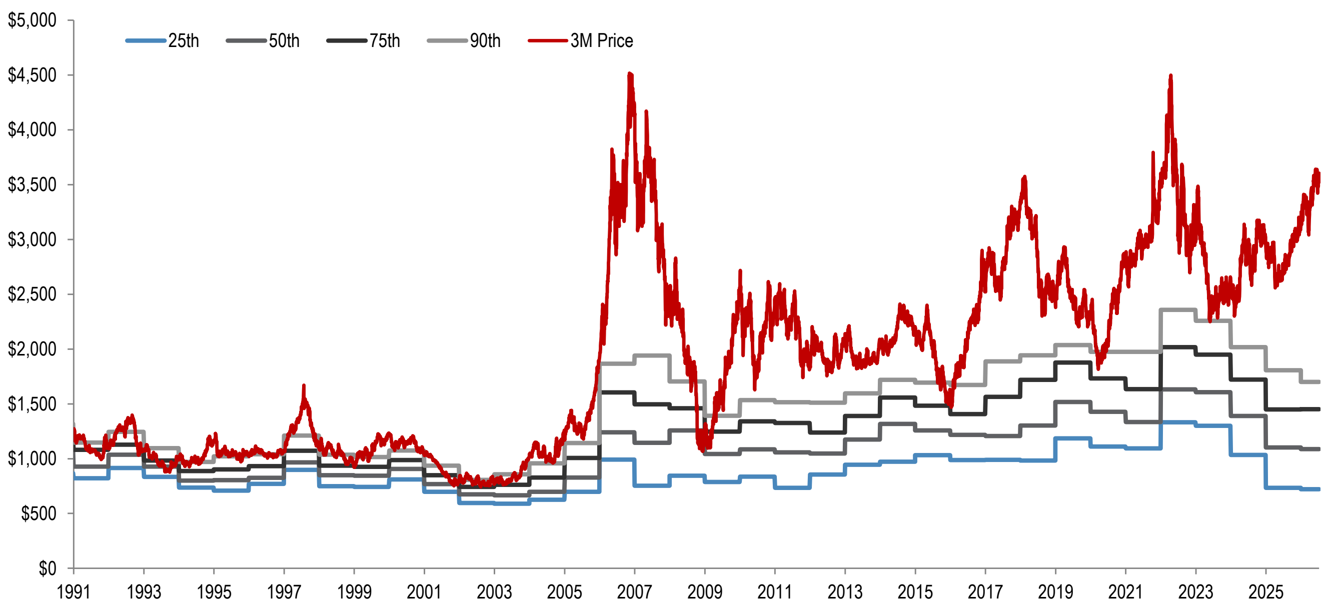


Source: SMM, J.P. Morgan Commodities Research

Source: .

Figure 88: Global zinc C1 mining costs by %-tile and LME 3M copper price

US$/mt MOD



Source: Wood Mackenzie, LME, J.P. Morgan Commodities Research. *LME prices as of 9th July 2026.

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Table 7: China zinc trade flow

Thousand mt; Percent Change, yoy.

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Concentrate Imports (Gross)	2,969	3,183	3,822	3,648	4,135	4,723	4,097	5,331	2,399
Refined Zinc Imports	715	605	541	434	79	380	446	304	54
Unwrought Zinc and Alloys Imports	821	692	615	520	142	429	495	348	67
Refined Zinc Exports	22	62	29	5	81	9	17	94	18
Unwrought Zinc and Alloys Exports	24	64	32	7	83	12	22	101	18
Net Refined Imports	693	544	512	429	-2	371	429	210	36
YoY-%									
Concentrate Imports (Gross)		7.2%	20.1%	-4.6%	13.3%	14.2%	-13.3%	30.1%	8.8%
Refined Zinc Imports		-15.4%	-10.6%	-19.8%	-81.8%	380.1%	17.2%	-31.8%	-65.3%
Unwrought Zinc and Alloys Imports		-15.7%	-11.2%	-15.4%	-72.6%	201.7%	15.3%	-29.8%	-61.5%
Refined Zinc Exports		181.3%	-52.5%	-81.8%	1415.5%	-89.0%	88.2%	460.0%	75.5%
Unwrought Zinc and Alloys Exports		163.9%	-50.8%	-79.0%	1154.1%	-85.1%	78.4%	358.4%	43.4%

*2026 is through May.
Source: China Customs General Administration

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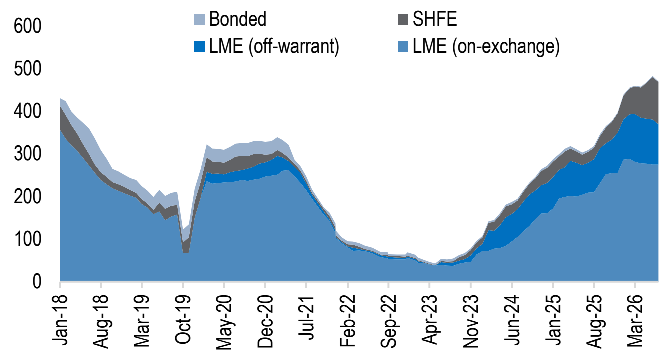
Nickel

Figure 89: Global visible nickel inventory

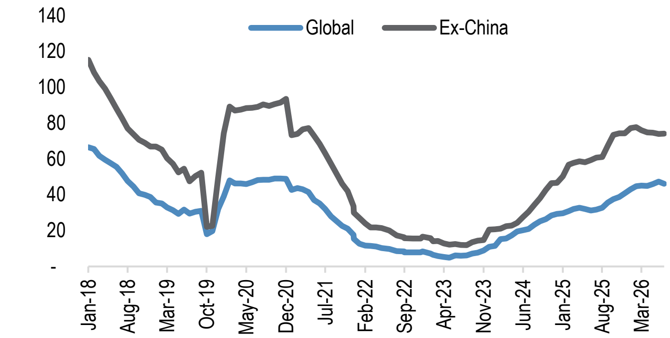
Thousand mt



Source: LME, SHFE, COMEX. Bonded warehouses are in China. As of 10th July 2026.

Figure 90: Global visible nickel inventory in days of use

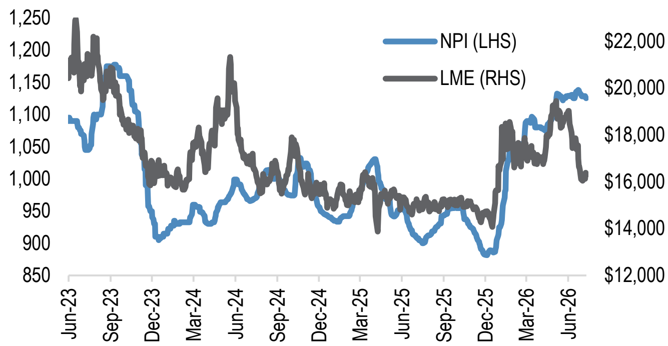
Days



Source: SHFE, J.P. Morgan Commodities Research

Figure 91: Chinese nickel pig iron prices (8-12%) and LME cash nickel prices

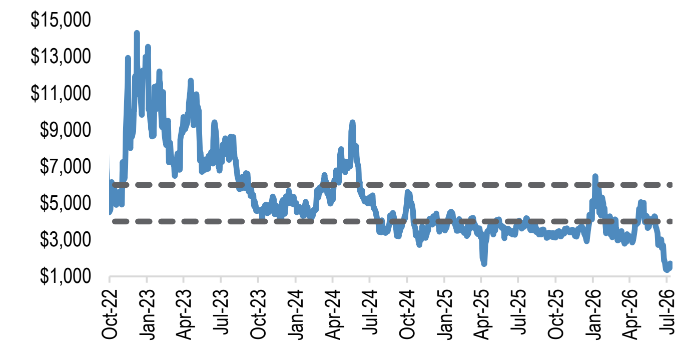
LHS: CNY/nickel tonne unit; RHS: US$/mt



Source: SMM, LME, Bloomberg Finance L.P., J.P. Morgan Commodities Research. As of 10th July 2026.

Figure 92: Discount spread of NPI prices compared to LME cash nickel prices

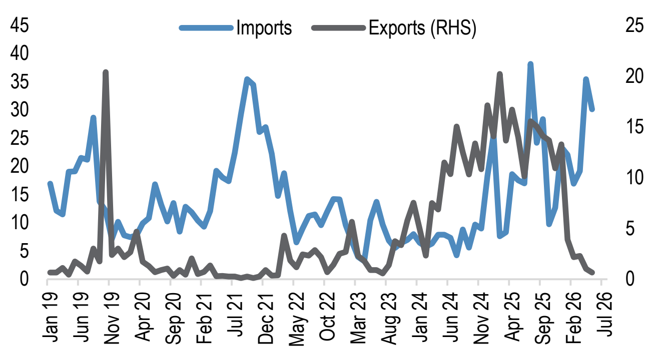
US$/mt. Dashed lines indicates level of estimated conversion costs



Source: SMM, LME, Bloomberg Finance L.P., J.P. Morgan Commodities Research. As of 10th July 2026.

Figure 93: China monthly refined nickel imports and exports

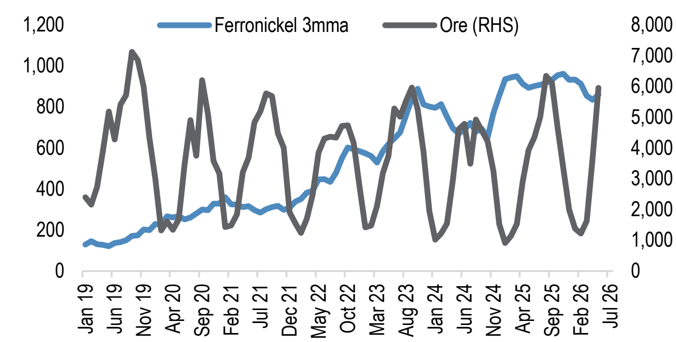
Thousand mt



Source: China Customs, Bloomberg Finance L.P.

Figure 94: China monthly ferronickel and nickel ore imports

Thousand mt



Source: China Customs, Bloomberg Finance L.P.

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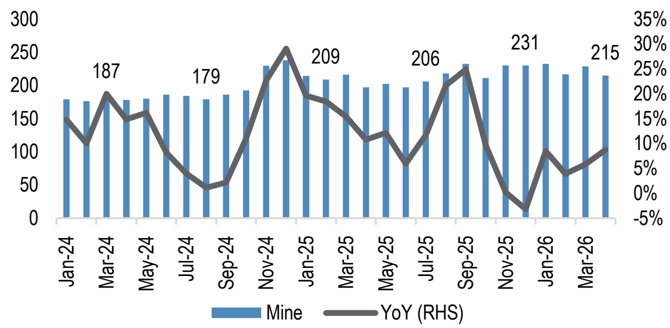
Figure 95: Indonesia monthly nickel mine production

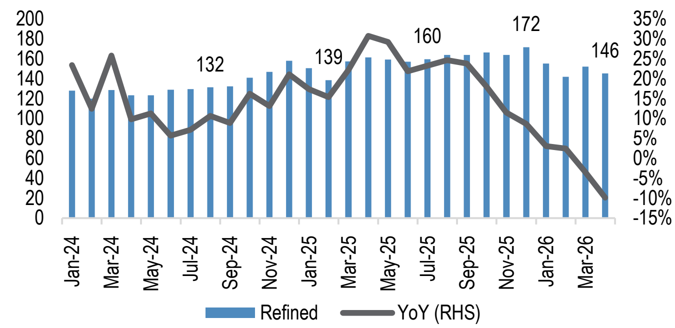
LHS: Thousand mt; RHS: Percent change, yoy



Source: Bloomberg Finance L.P.

Figure 96: Indonesia monthly refined nickel production

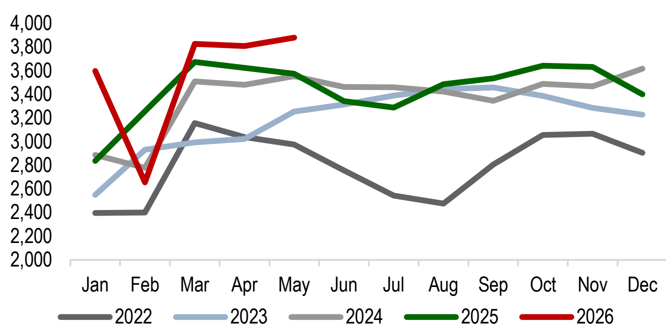
LHS: Thousand mt; RHS: Percent change, yoy



Source: Bloomberg Finance L.P.

Figure 97: China stainless steel production

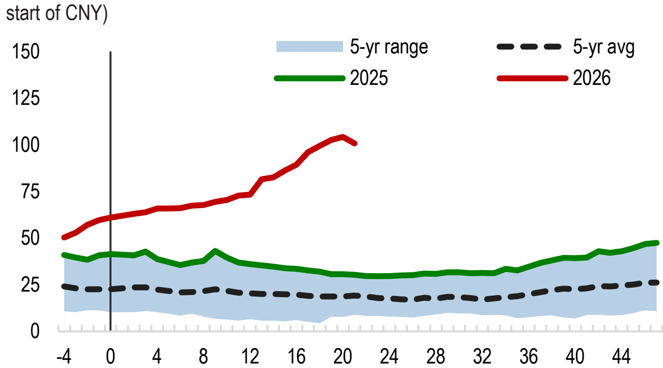
Thousand mt



Source: CRU

Figure 98: Total visible China nickel inventory (SHFE + Regional Warehouses)

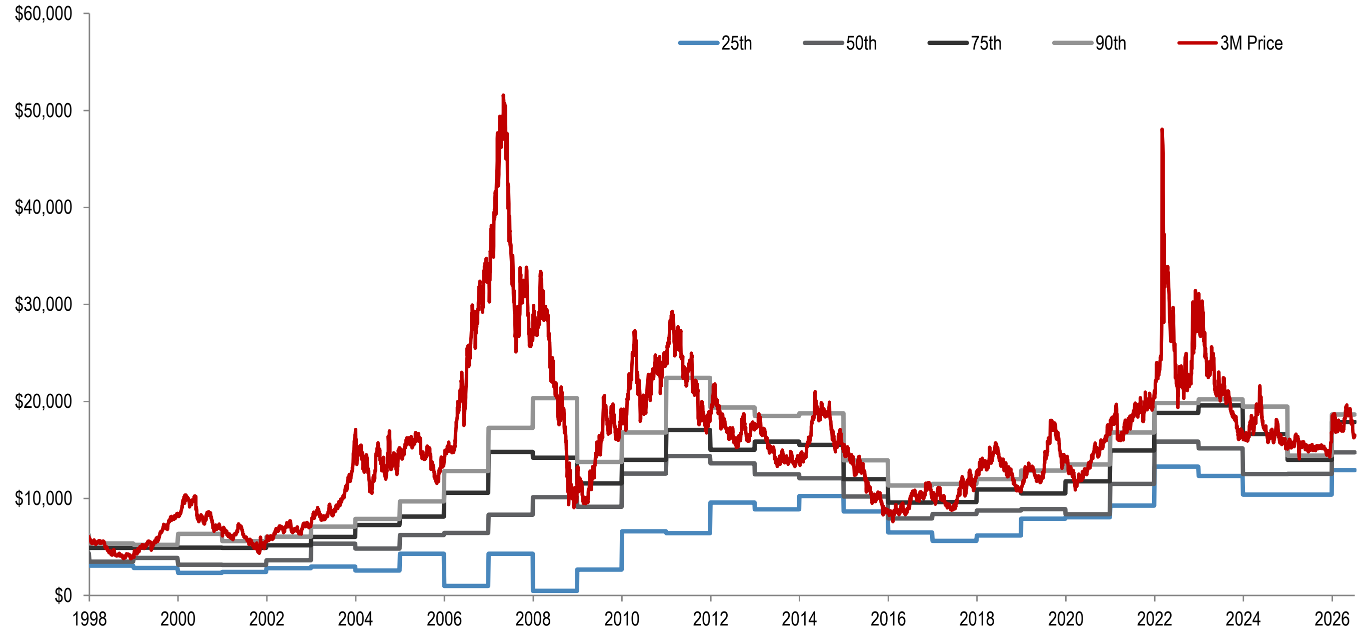
Y-axis: Thousand mt; X-axis: Weeks around Chinese New Year (0 = week closest to start of CNY)



Source: SHFE, SMM, J.P. Morgan Commodities Research

Figure 99: Global nickel industry C1 costs by %-tile and LME 3M nickel price

US$/mt MOD



Source: Wood Mackenzie, LME, J.P. Morgan Commodities Research. *LME prices as of 9th July 2026.

Table 8: China nickel trade flow

Thousand mt; Percent Change, yoy

	2018	2019	2020	2021	2022	2023	2024	2025	2026 YTD
Nickel Ore Imports	46,971	56,116	39,104	43,529	40,241	44,819	38,096	42,263	13,863
Refined Nickel Imports	212	193	131	261	156	91	97	231	124
Refined Nickel Exports	13	37	19	5	23	37	123	171	10
Net Refined Imports	199	156	112	256	133	54	-25	60	114
YoY-%									
Nickel Ore Imports		19.5%	-30.3%	11.3%	-7.6%	11.4%	-15.0%	10.9%	32.9%
Refined Nickel Imports		-8.9%	-32.3%	99.8%	-40.3%	-41.6%	6.8%	137.7%	59.4%
Refined Nickel Exports		182.5%	-49.3%	-71.3%	314.2%	62.7%	235.0%	39.7%	-87.2%

*2026 is through May.
Source: China Customs General Administration

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JPM metals price forecasts

Quarterly and annual averages

	4Q25A	1Q26A	2Q26A	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27	2025	2026	2027
Base Metals												
Aluminum	2,828	3,198	3,562	3,800	3,700	3,450	3,200	2,900	2,750	2,631	3,565	3,075
Copper	11,105	12,824	13,352	14,500	14,800	14,000	13,800	13,800	13,600	9,947	13,869	13,800
Nickel	14,892	17,338	18,090	17,000	17,250	17,500	18,000	18,000	17,500	15,164	17,419	17,750
Zinc	3,165	3,237	3,464	3,500	3,400	3,300	3,200	3,050	3,000	2,867	3,400	3,138
Precious Metals												
Gold	4,152	4,873	4,508	4,300	4,500	4,600	4,700	4,800	5,000	3,440	4,545	4,775
Silver	55.2	83.7	73.1	62.5	63.0	63.0	63.5	64.0	65.0	40.1	70.6	63.9
Platinum	1,694	2,204	1,916	1,700	1,800	1,875	1,900	1,950	1,950	1,282	1,905	1,919
Palladium	1,478	1,706	1,409	1,300	1,350	1,400	1,350	1,250	1,200	1,151	1,441	1,300

Source: J.P. Morgan Commodities Research. *Base metals are LME cash and precious metals are spot.

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